A Rational Neural Network Model of Memory Retrieval for Syntactic Dependency Resolution

William Timkey[1], Brian Dillon[2], Tal Linzen[1,3]

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[3] Center for Data Science, New York University

AMLaP 2026

Consider the following sentence:

Consider the following sentence:

The

Consider the following sentence:

resident

Consider the following sentence:

who

Consider the following sentence:

was

Consider the following sentence:

living

Consider the following sentence:

near

Consider the following sentence:

the

Consider the following sentence:

dangerous

Consider the following sentence:

neighbor

Consider the following sentence:

complained

Consider the following sentence:

about

Consider the following sentence:

the

Consider the following sentence:

investigation.

Consider the following sentence:

Who complained about the investigation?

a) The resident
b) The neighbor

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The resident who was living near the dangerous neighbor complained about the investigation.

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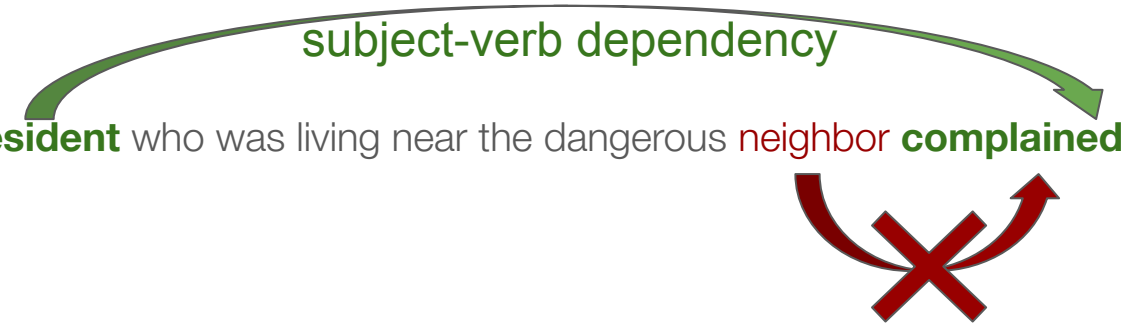
The **resident** who was living near the dangerous neighbor **complained** about the investigation.

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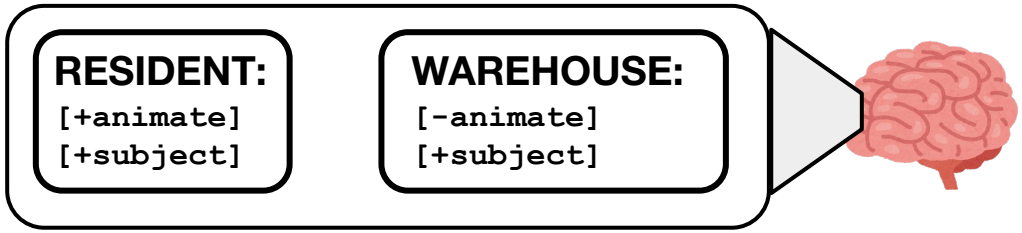
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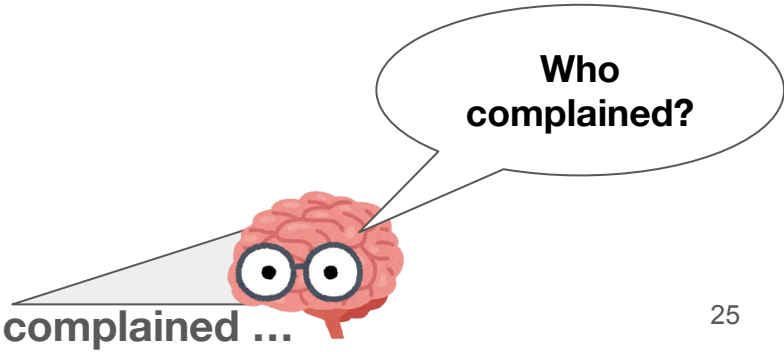


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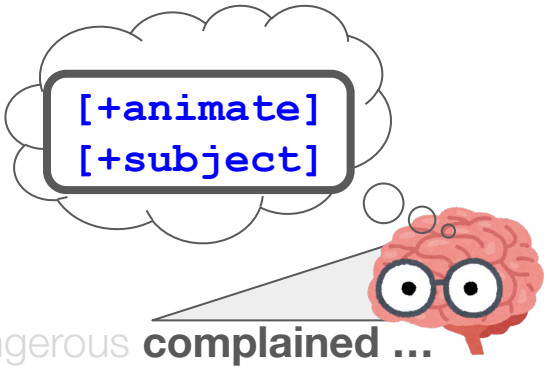


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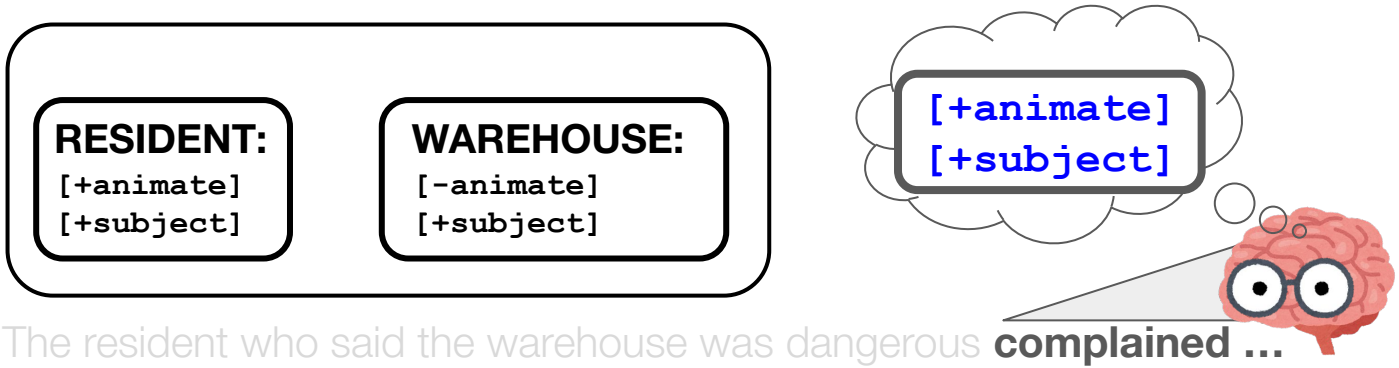
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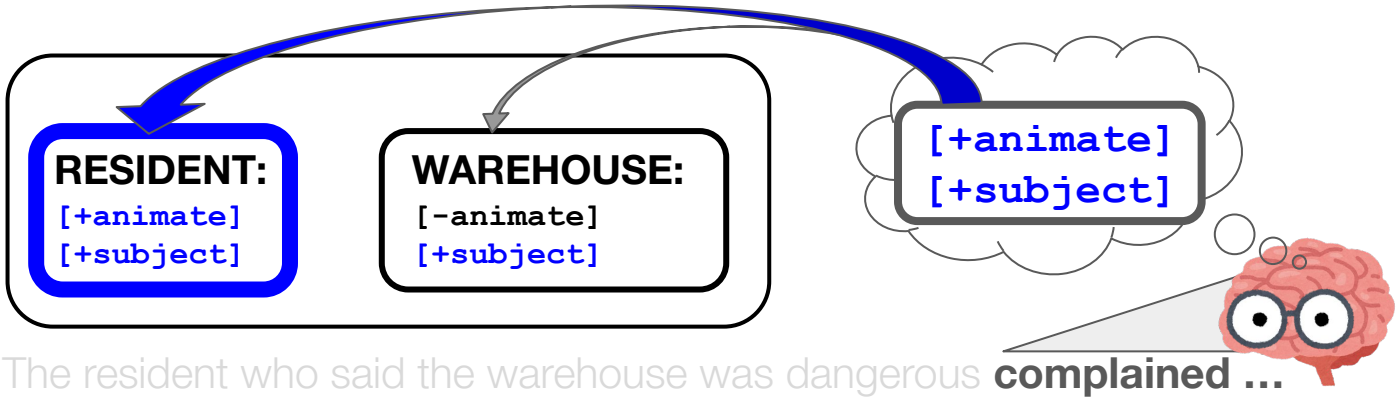
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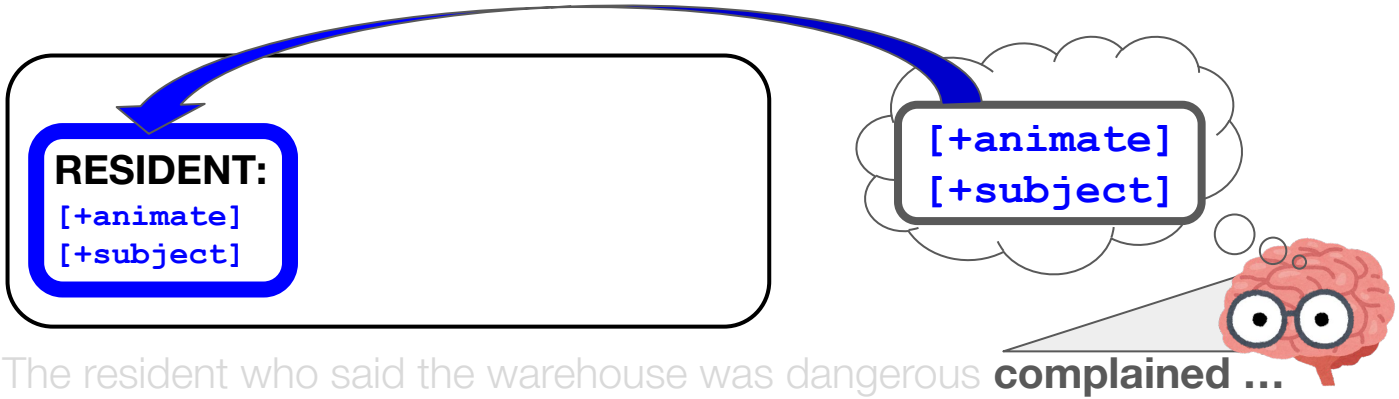
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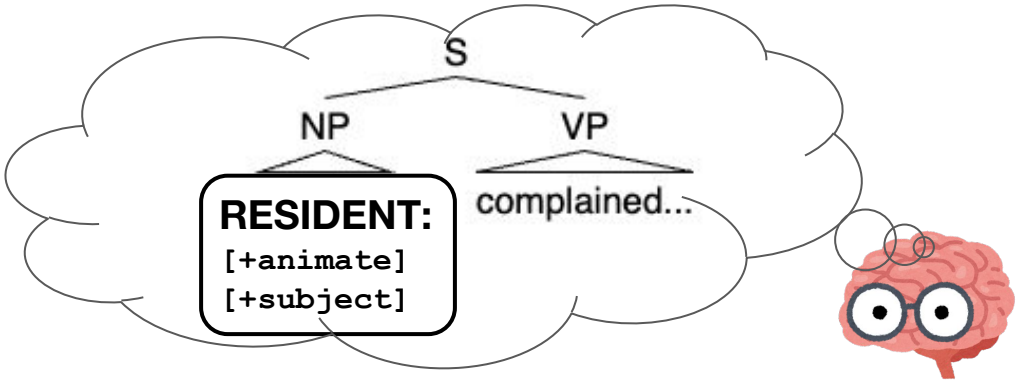
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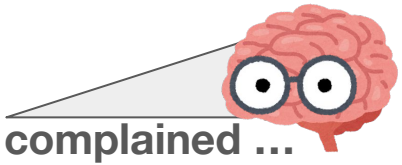
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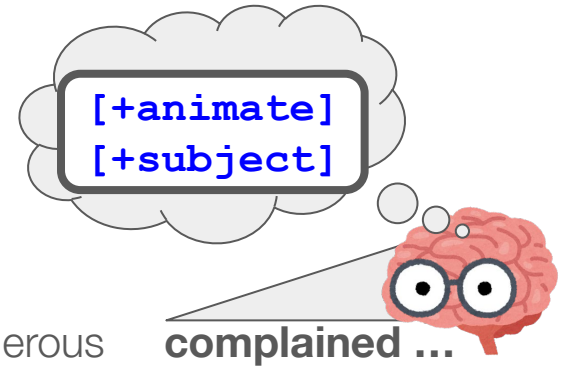
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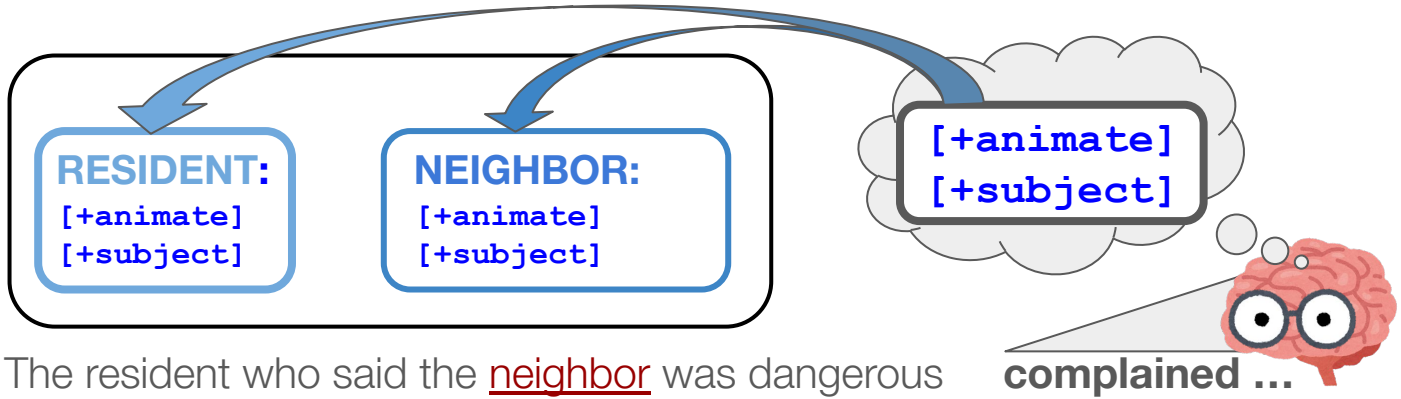
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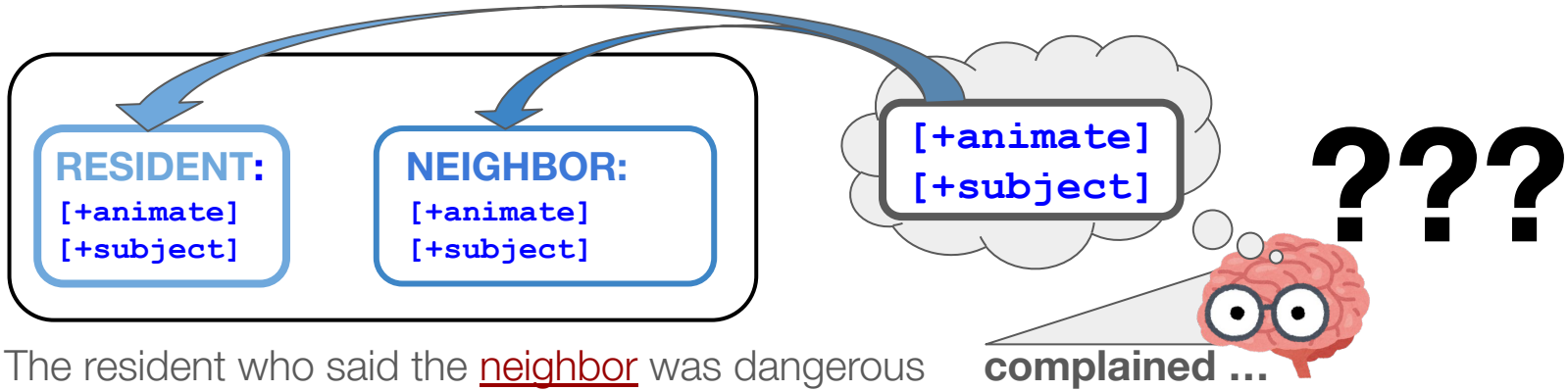
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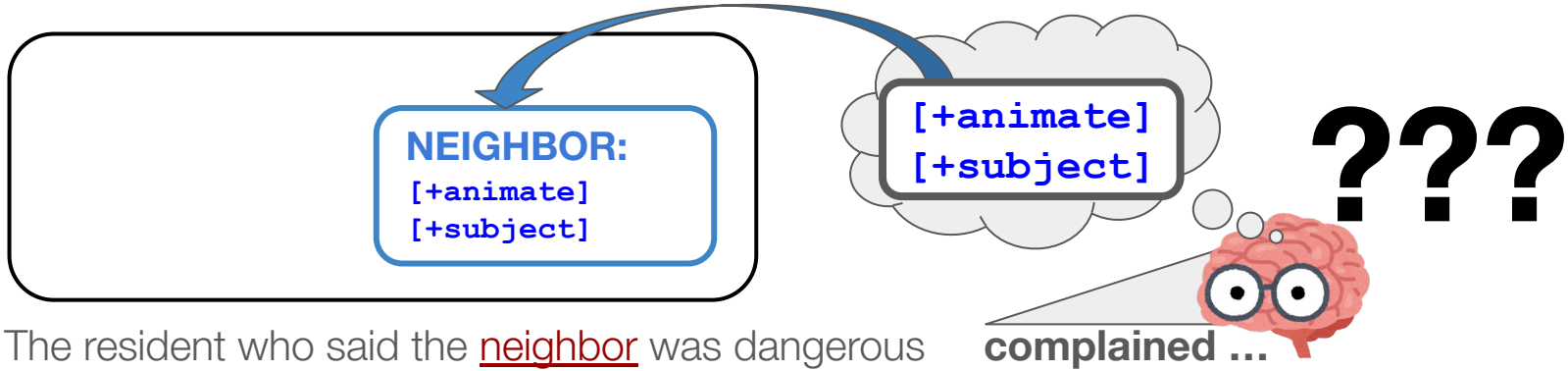
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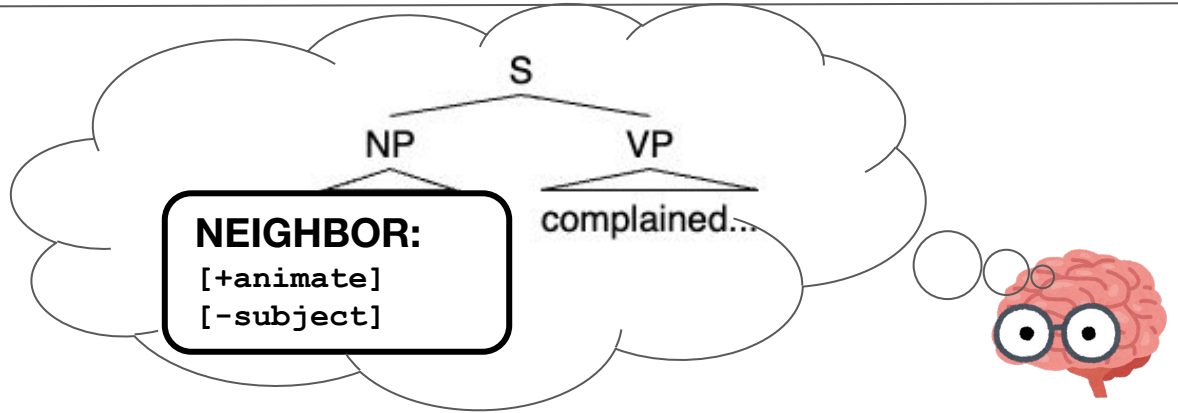
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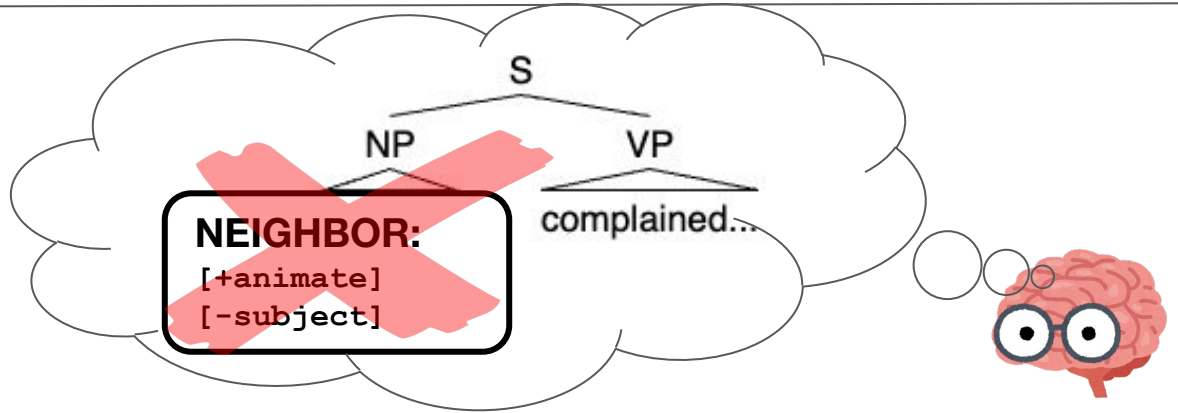
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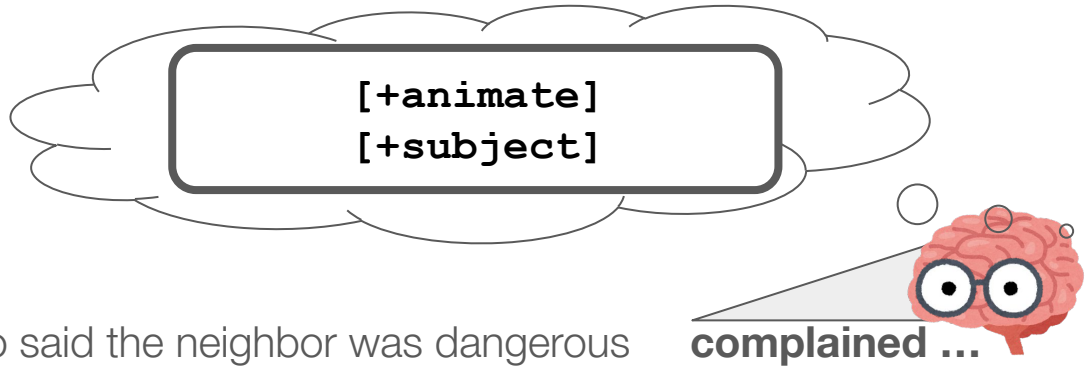
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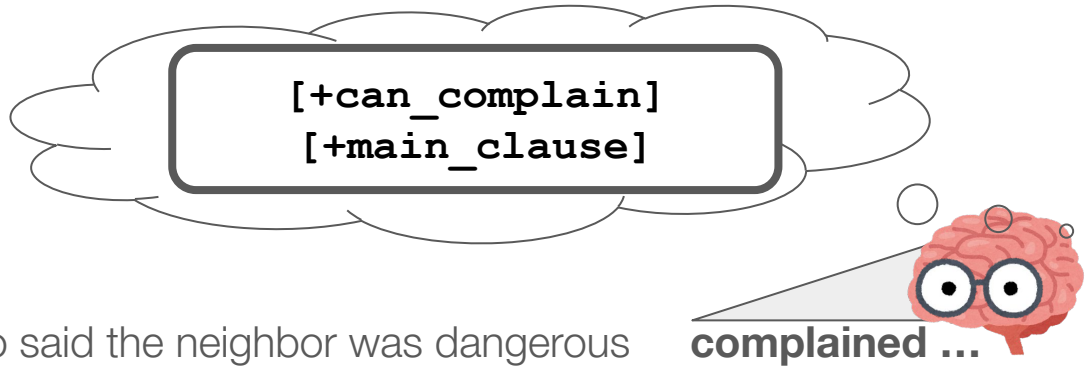
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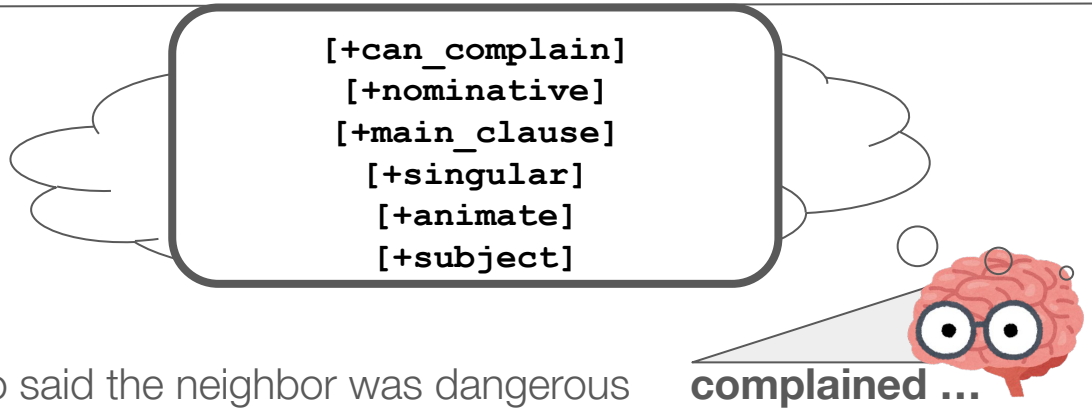
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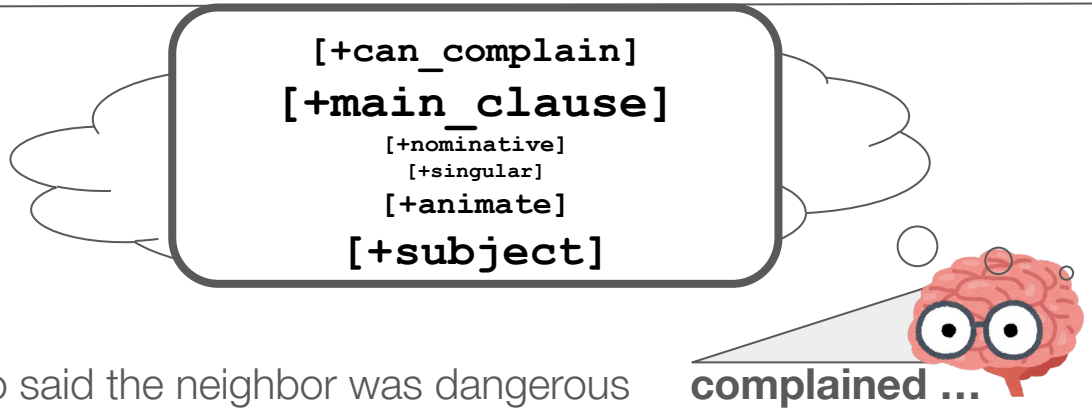
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  - semantic compatibility (Cunnings & Sturt 2018)
- How can we identify features that are plausibly used by humans given the vast space of possibilities?

A Rational approach

- Assume that human working memory is optimally adapted to the environment
- We learn representations that **maximize** our chances of retrieving the **correct** thing from working memory at the correct time, and **minimize interference** (Anderson 1991; 2004)

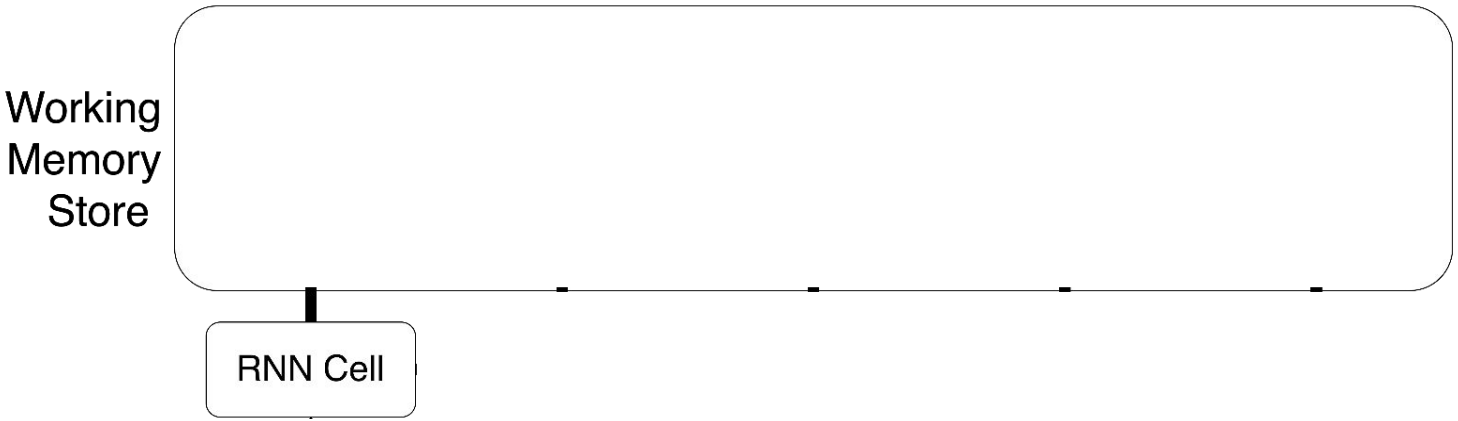
- New question: What representations would a system learn if its goal was to minimize interference when resolving syntactic dependencies?

A Rational approach

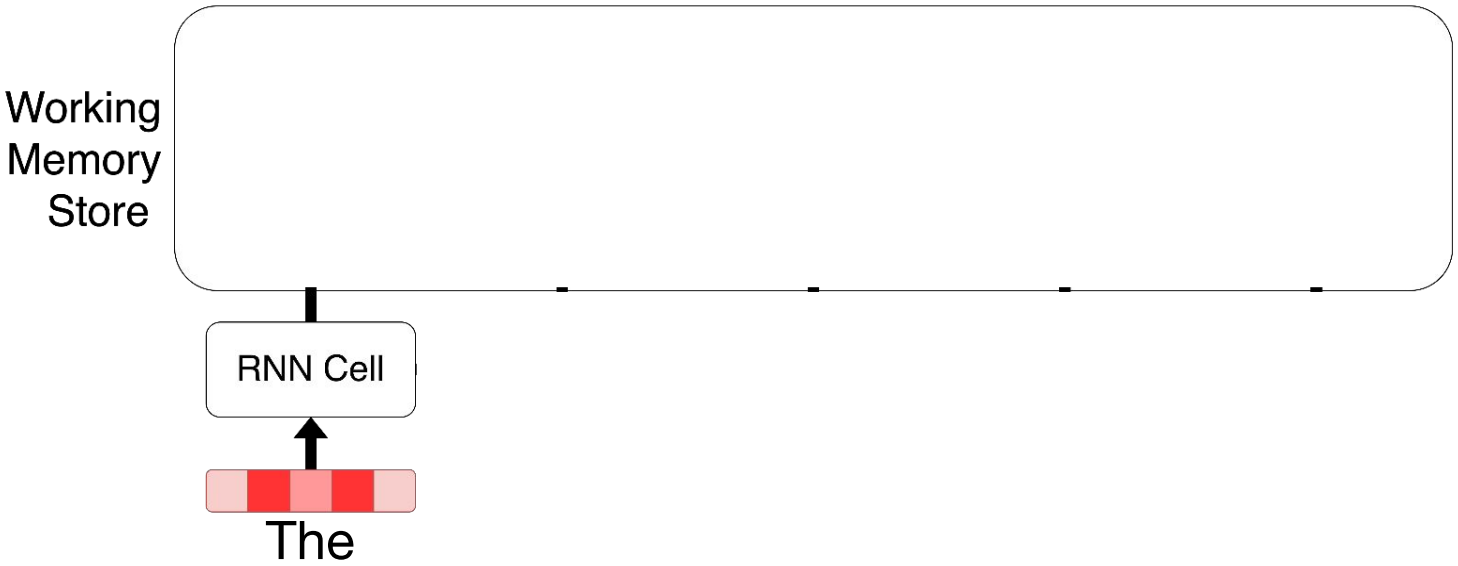
Our method:

1. Build a neural network dependency parser with a cue-based associative memory store
2. Initialize all parameters randomly
3. Train the model on next-word prediction and dependency resolution objective
4. Use NLP techniques to map the learned representations onto human-interpretable features
5. Test model-generated hypotheses in humans, iterate on model

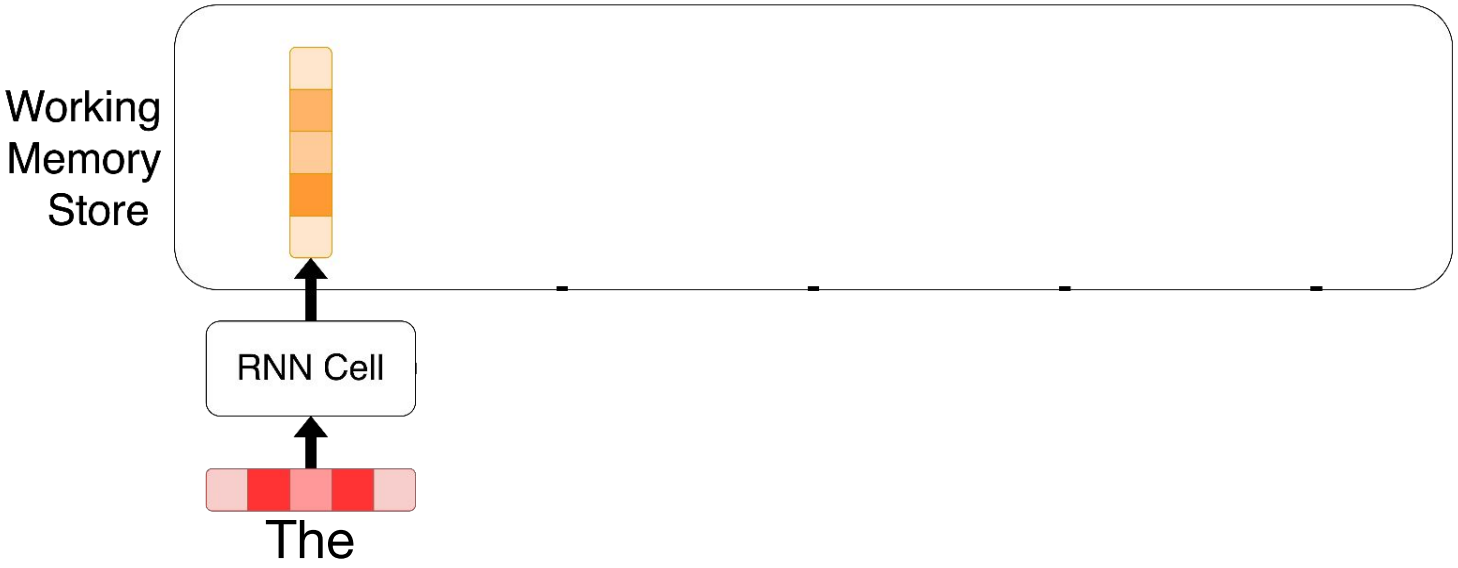
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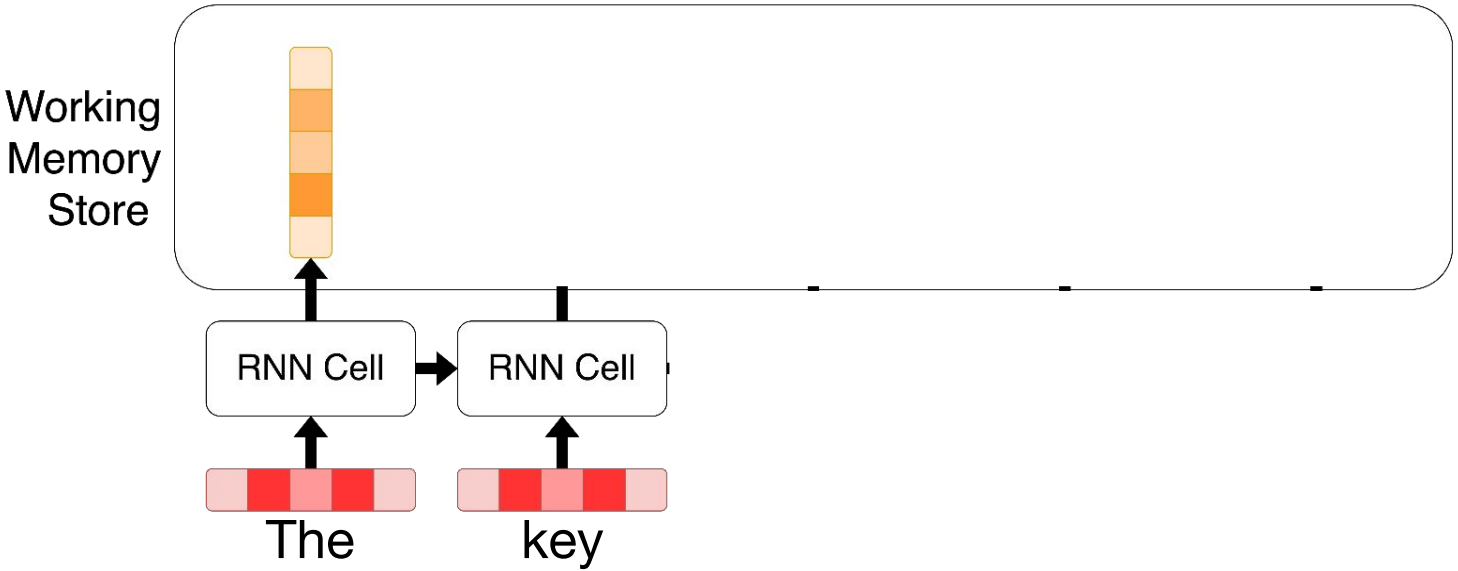
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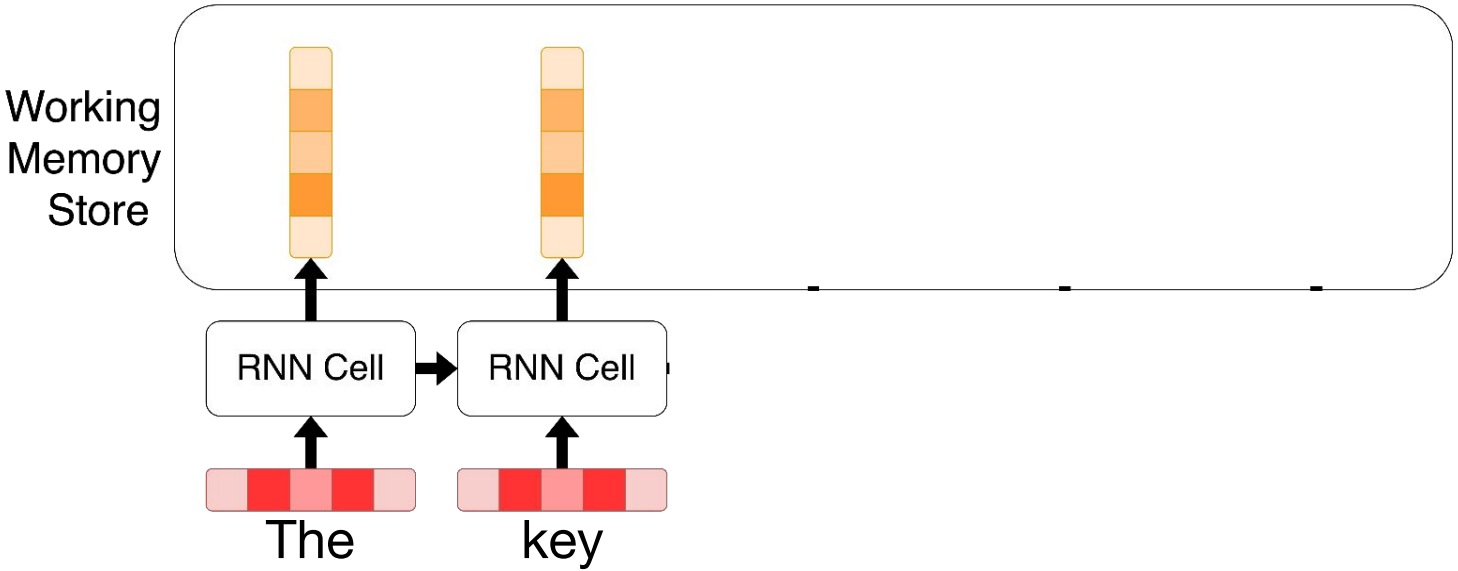
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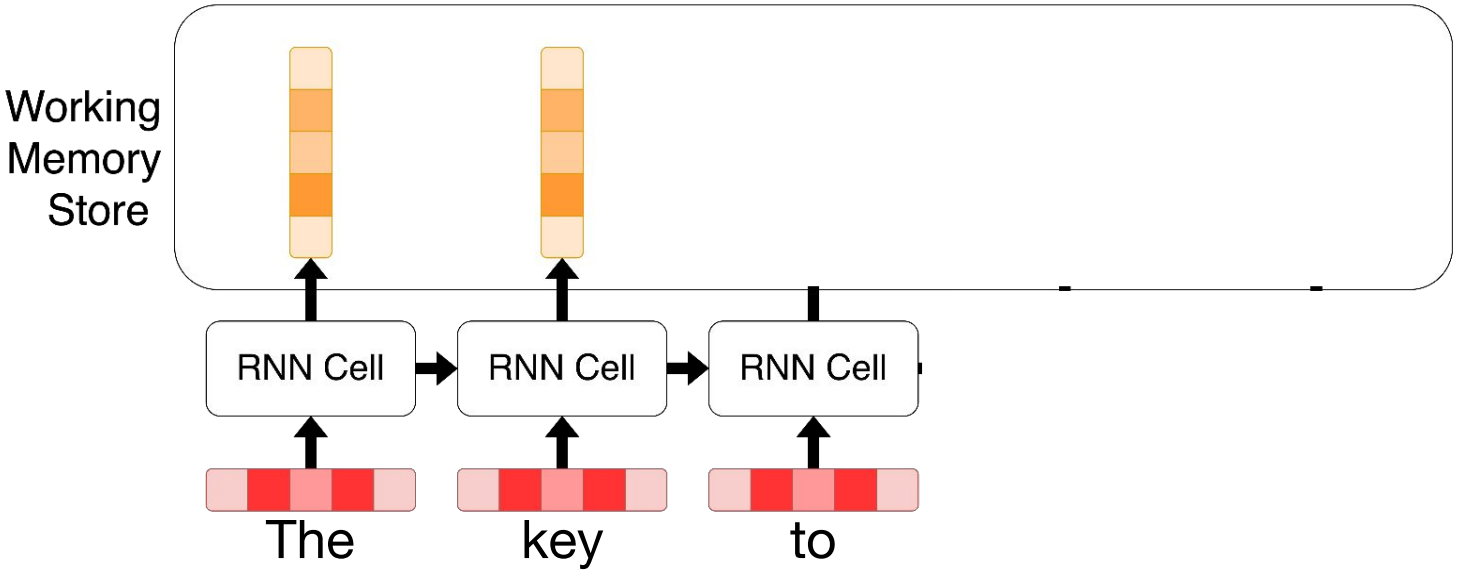
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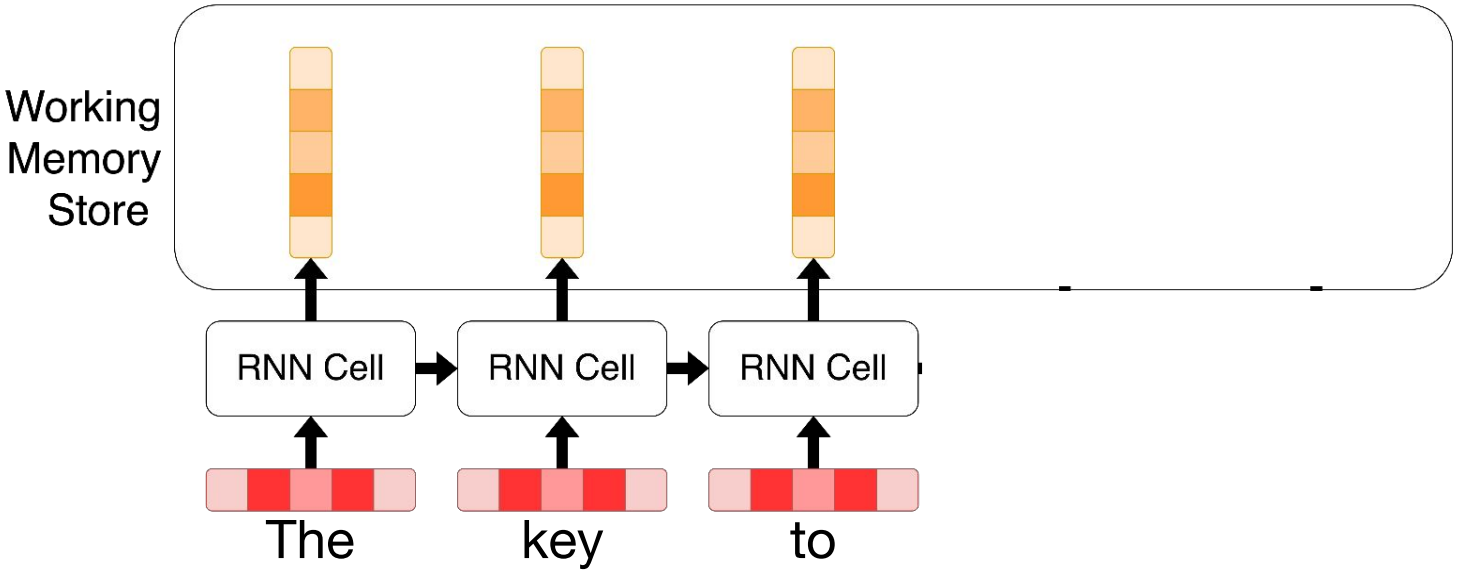
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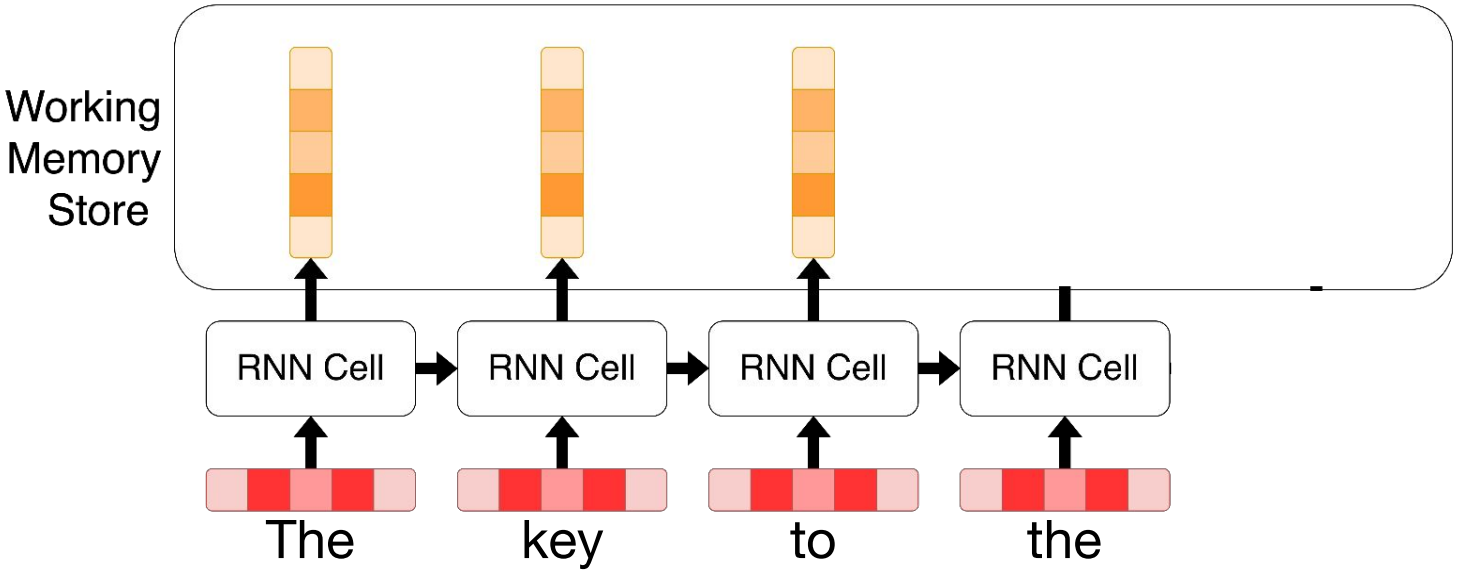
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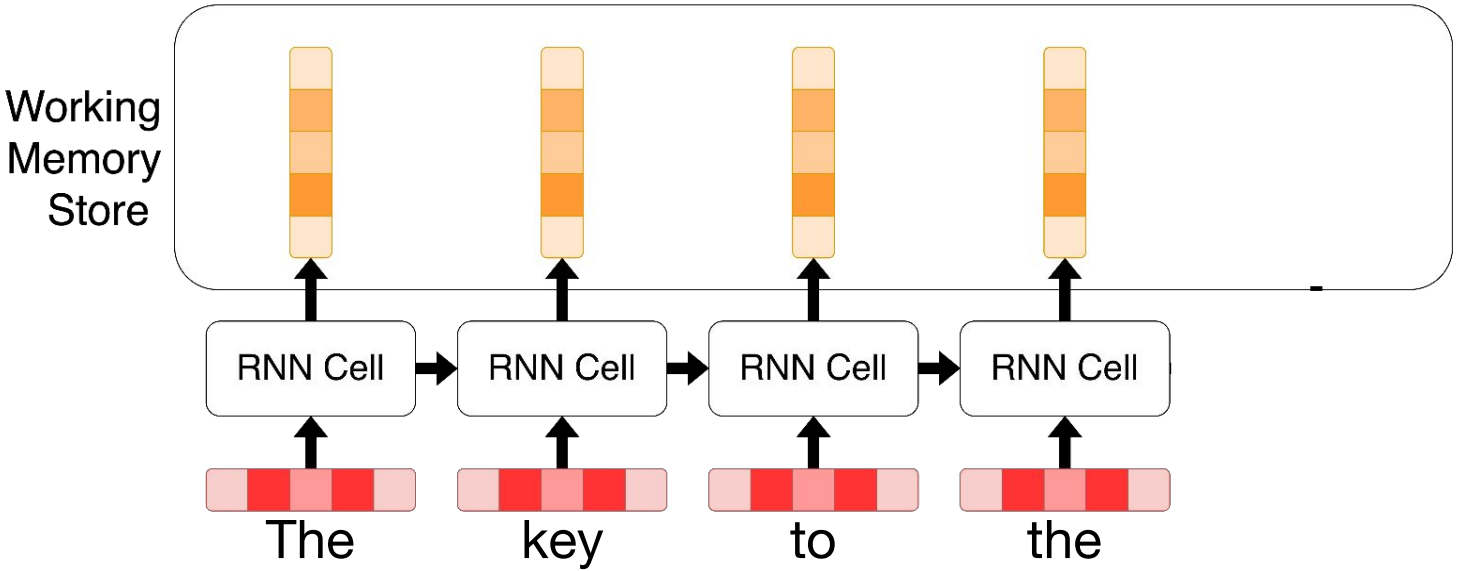
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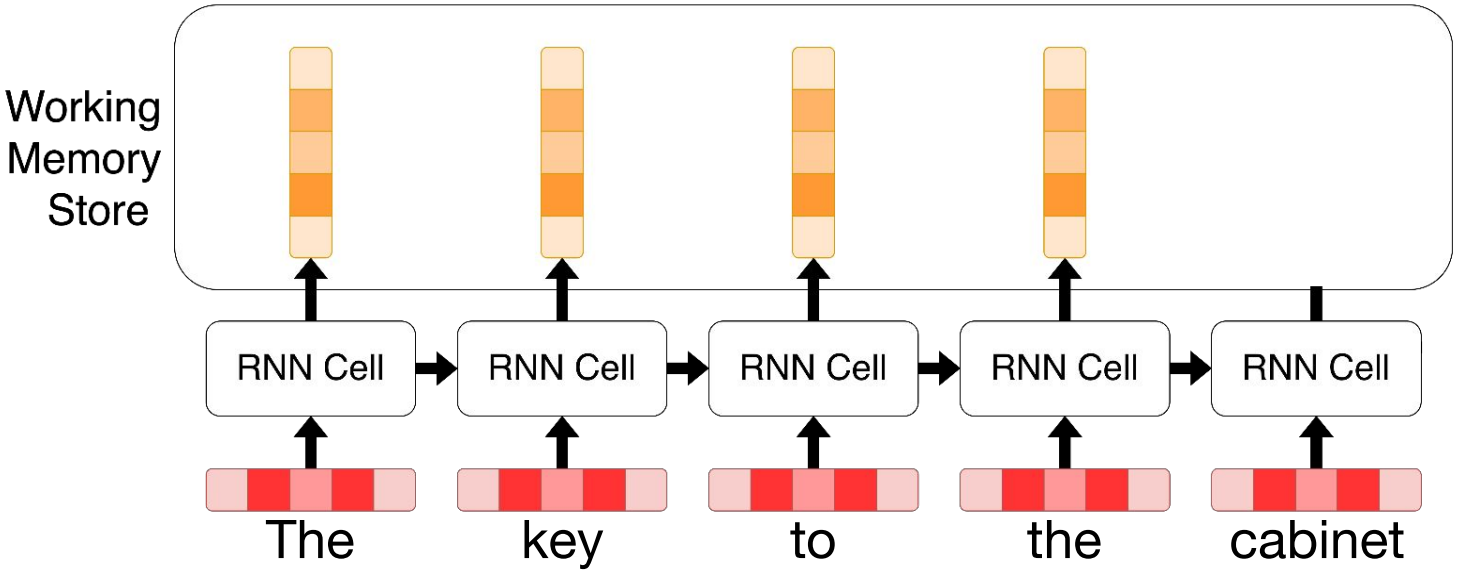
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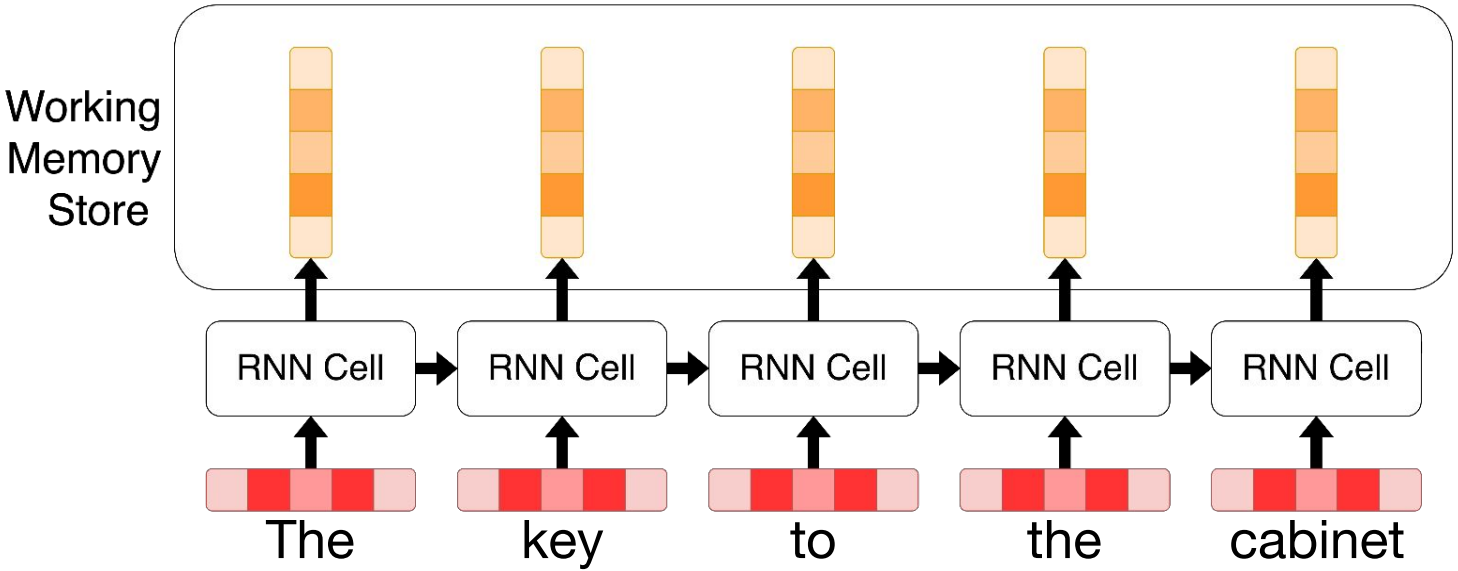
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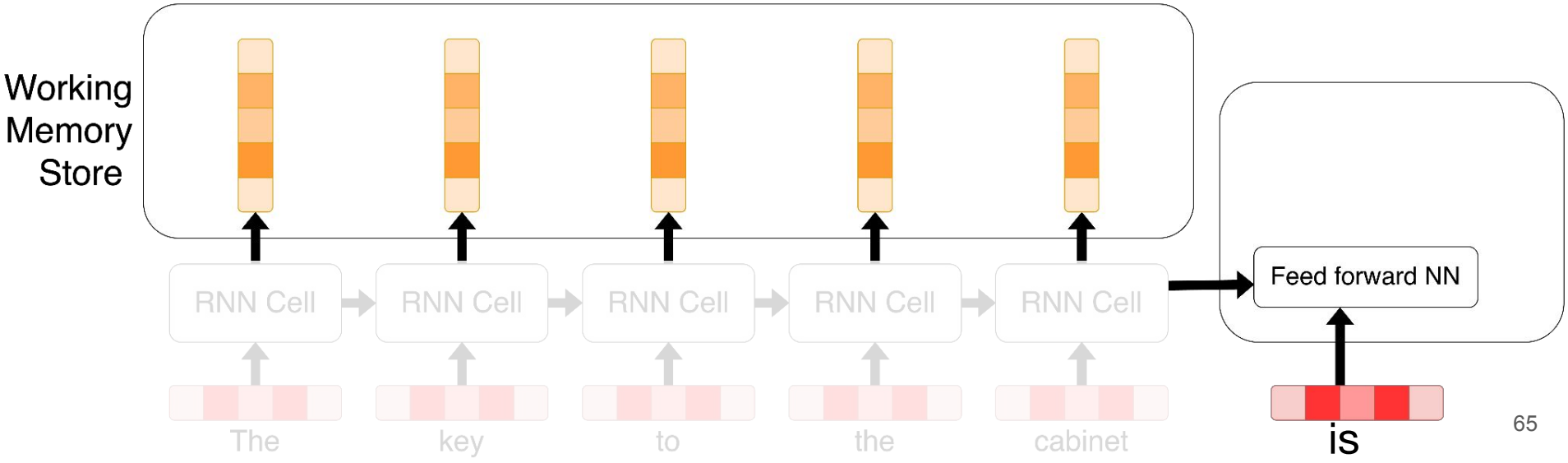
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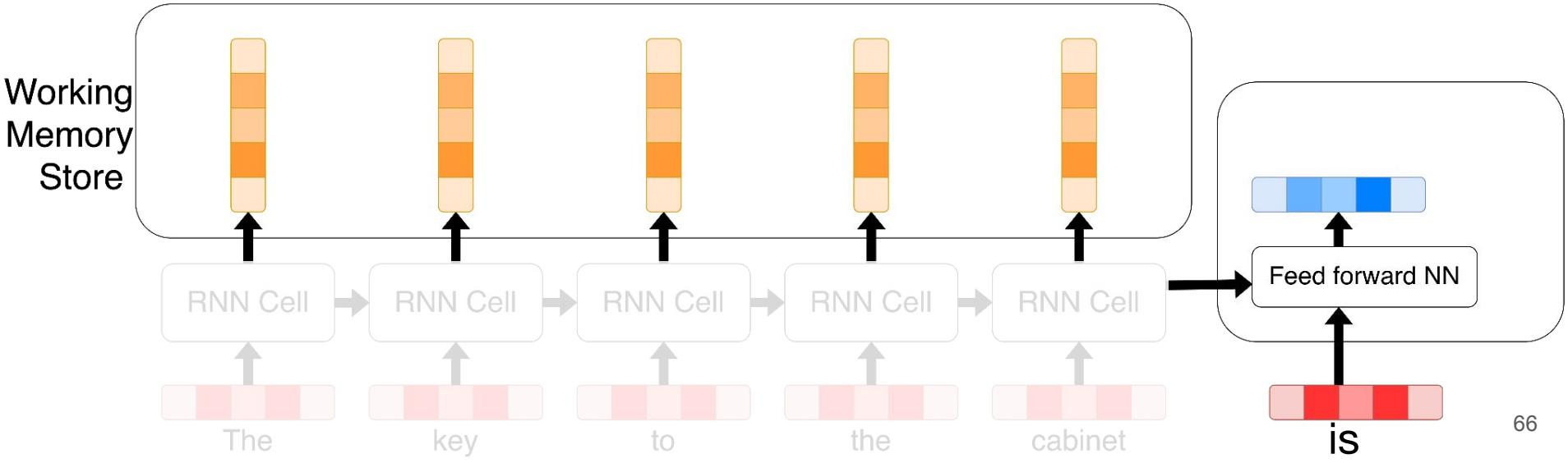
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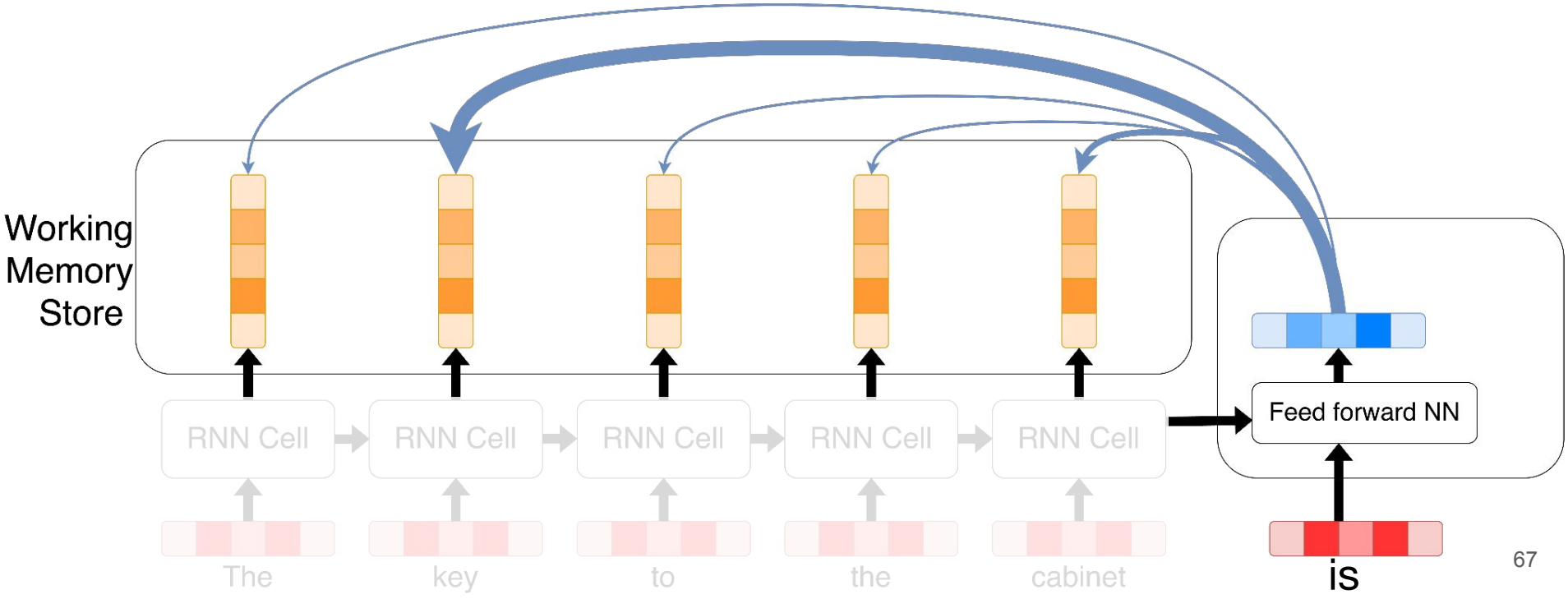
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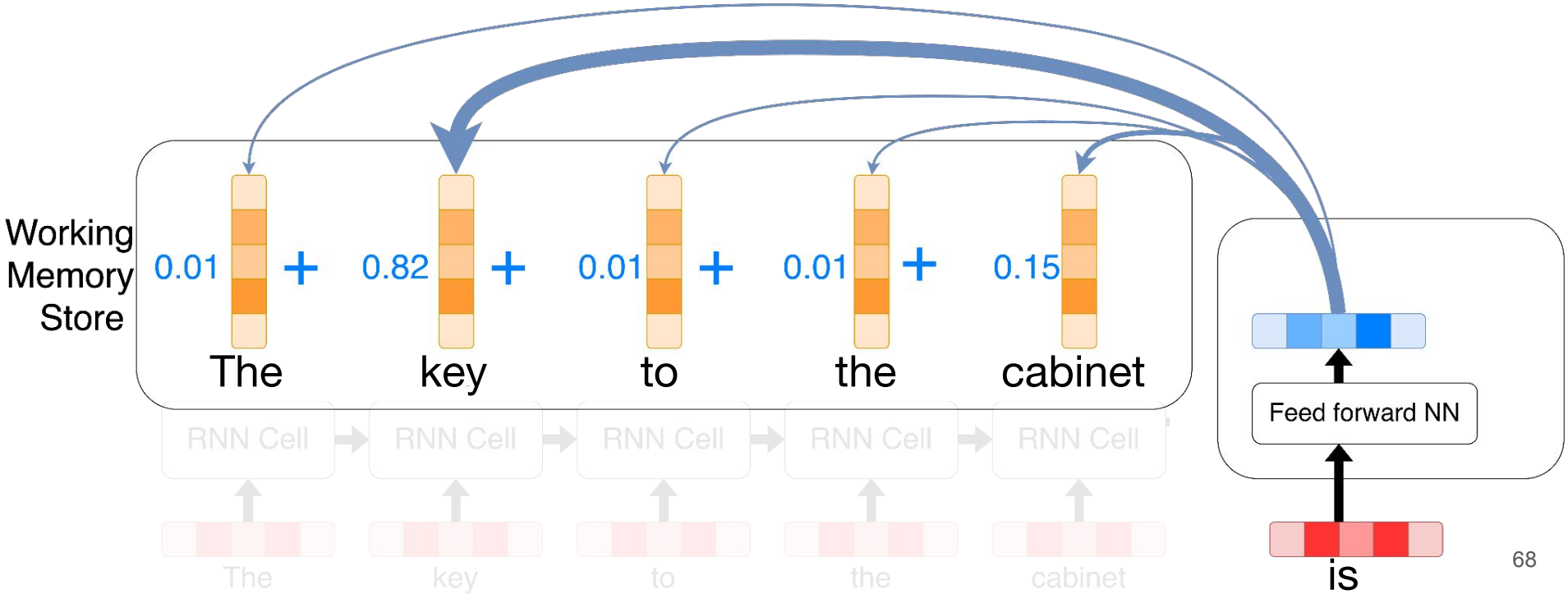
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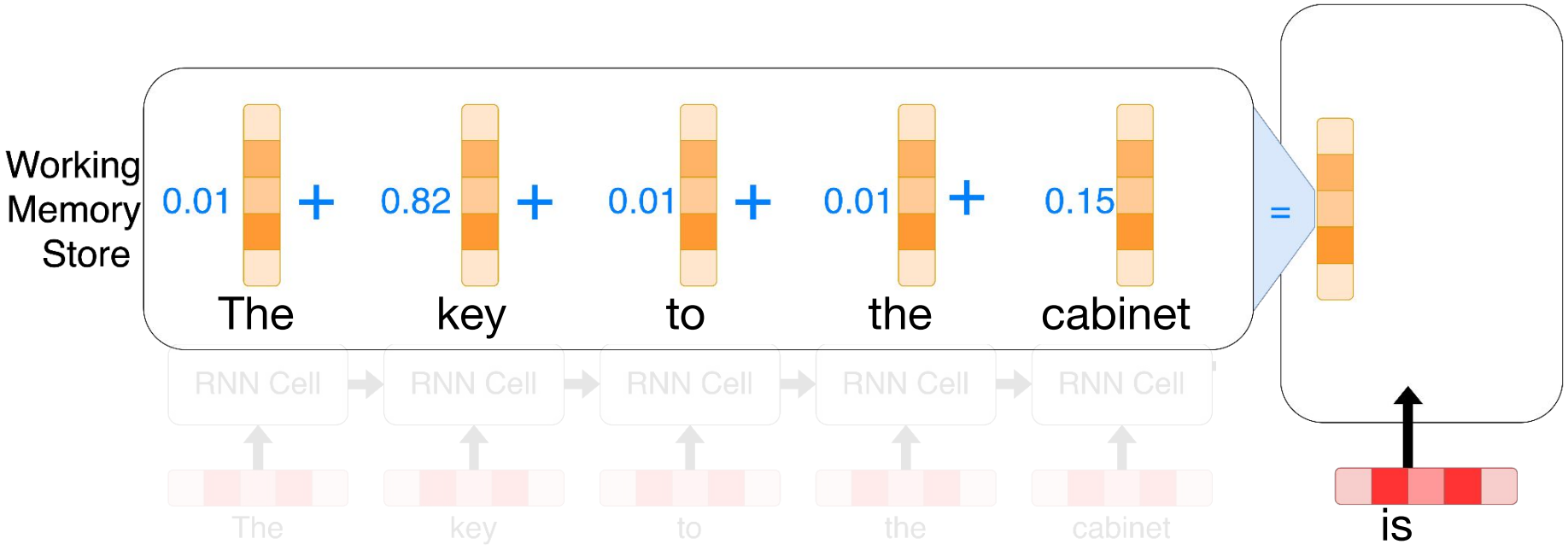
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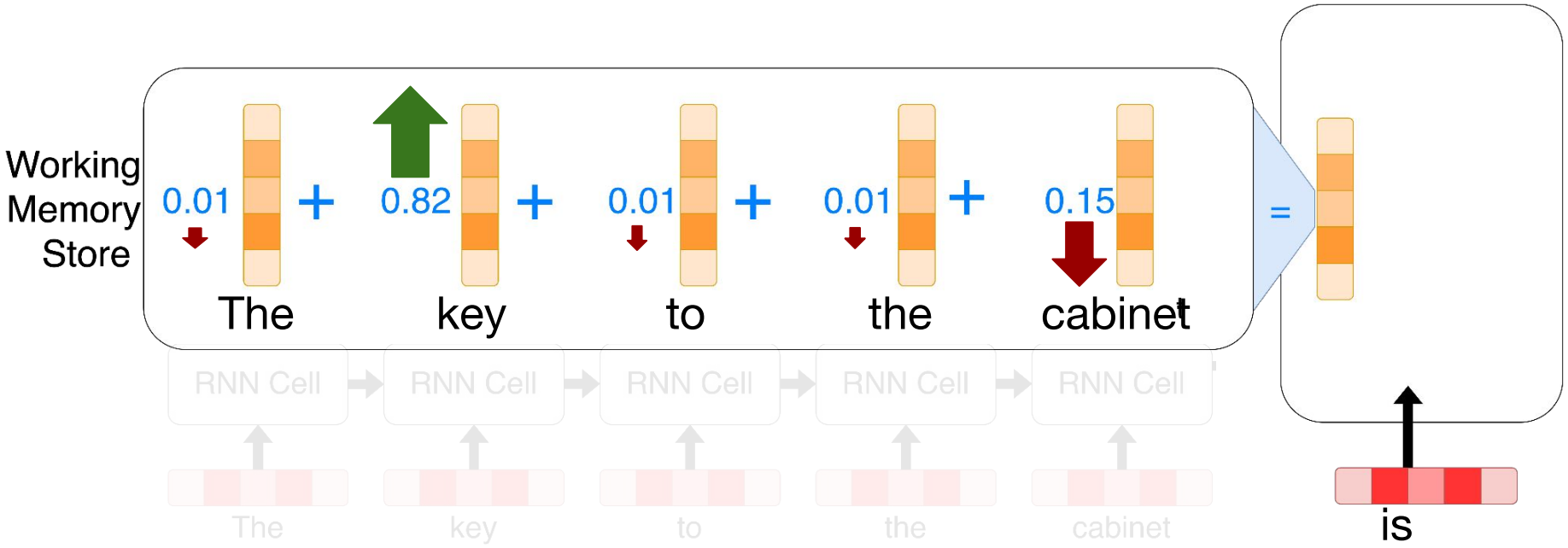
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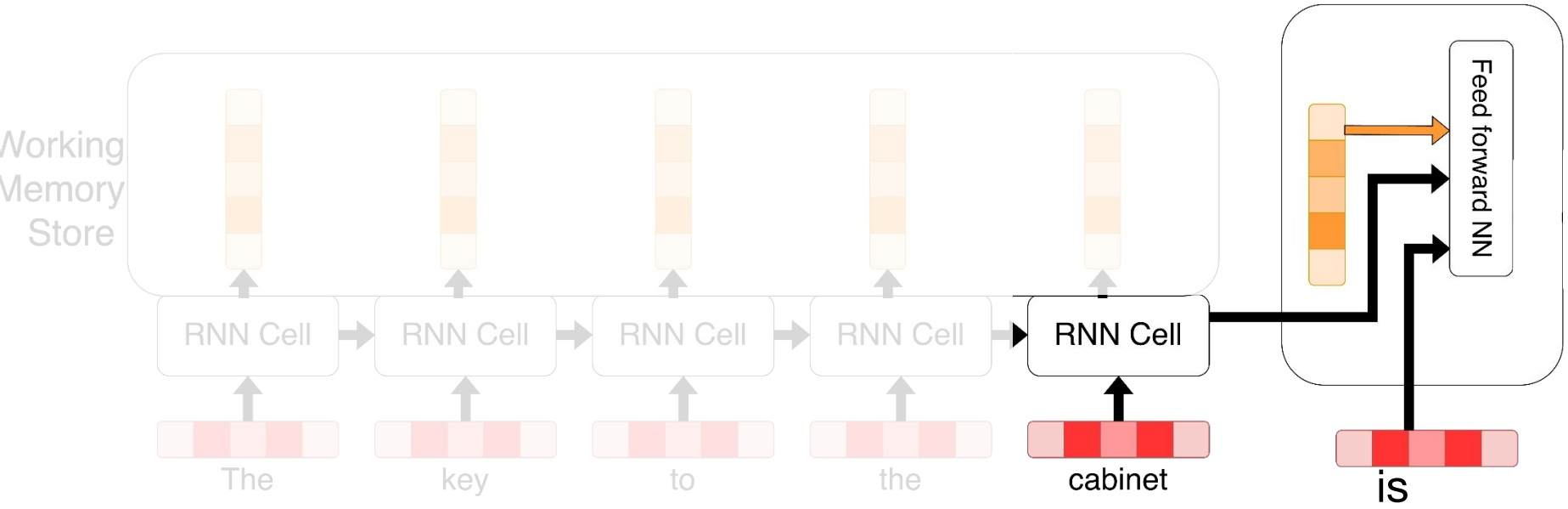
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Our model (simplified)

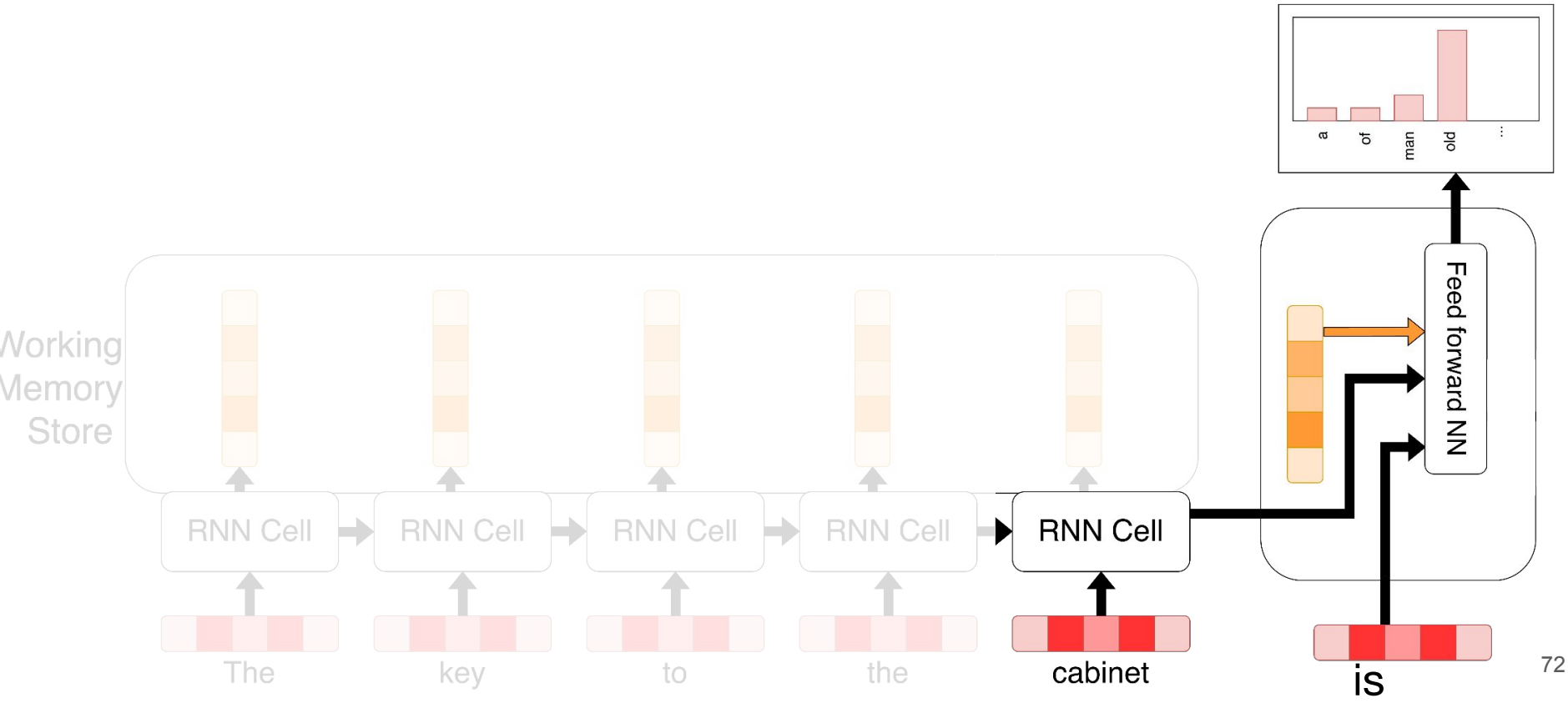


Our model (simplified)



Our model (simplified)

$$P(w_{i+1} \mid \textit{The key to the cabinet is})$$

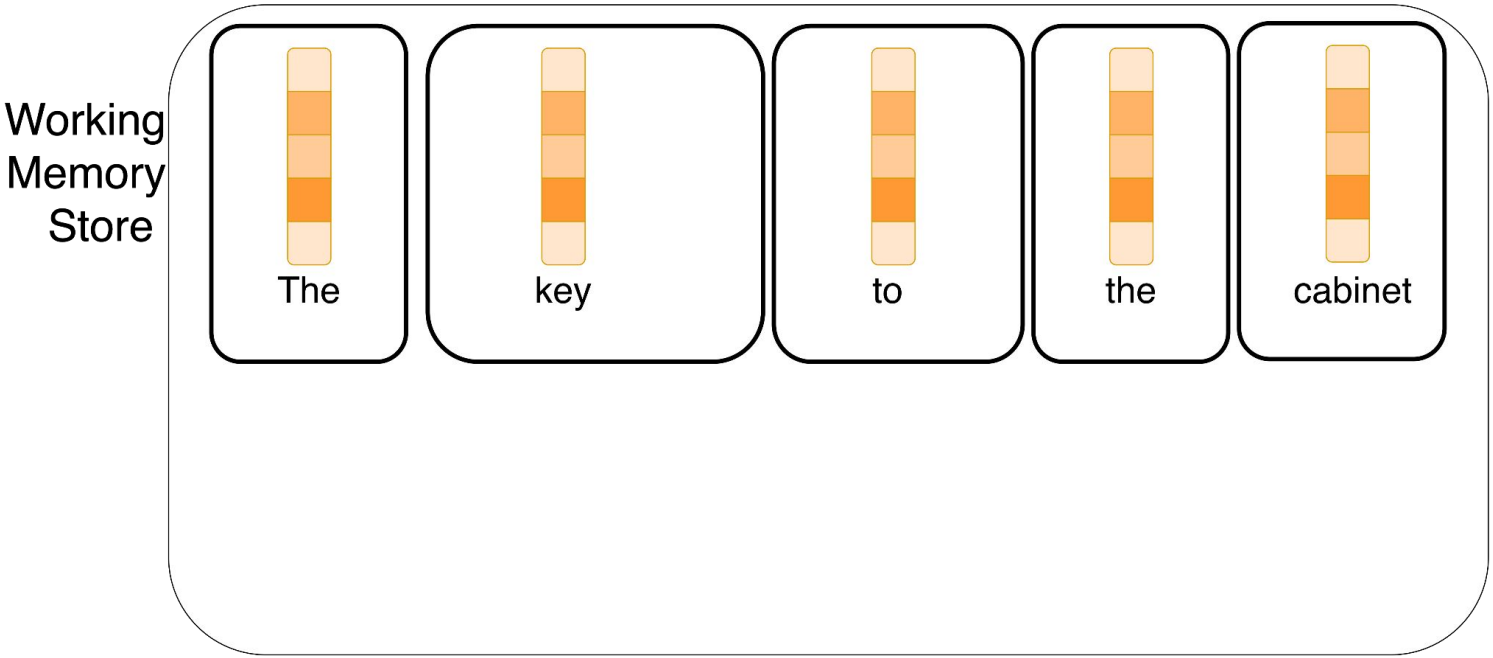


Actual model input:

```
SHIFT(The), SHIFT(key), LeftArc(), SHIFT(to), RightArc(),
SHIFT(the), SHIFT(cabinet), LeftArc(), RightArc()
```

Actual model input:

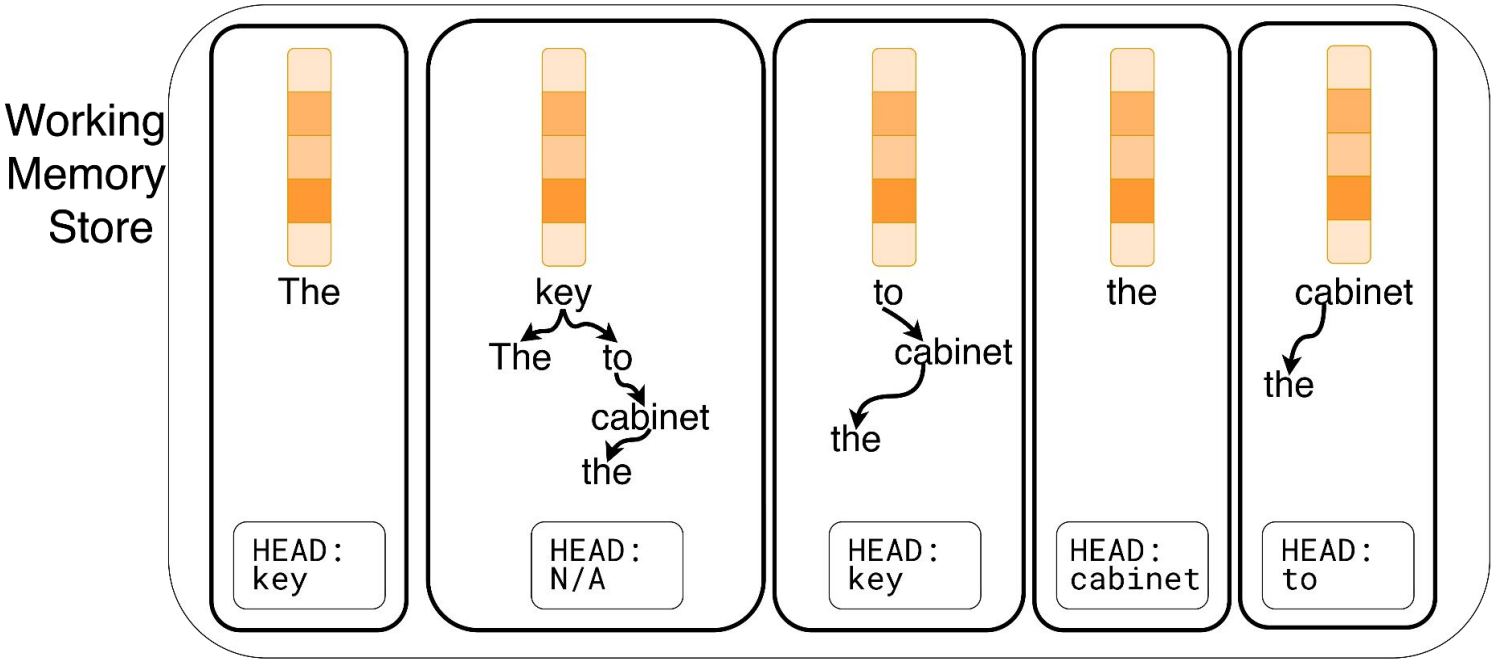
SHIFT(The), SHIFT(key), LeftArc(), SHIFT(to), RightArc(),
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Actual model input:

SHIFT(The), SHIFT(key), LeftArc(), SHIFT(to), RightArc(),
SHIFT(the), SHIFT(cabinet), LeftArc(), RightArc()

Actual model's working memory store:



Training objective

1. Predict next words and parsing actions
2. Retrieve the correct item from memory to resolve the current dependency

Training data: machine-parsed version of *wikitext-103* corpus (English)

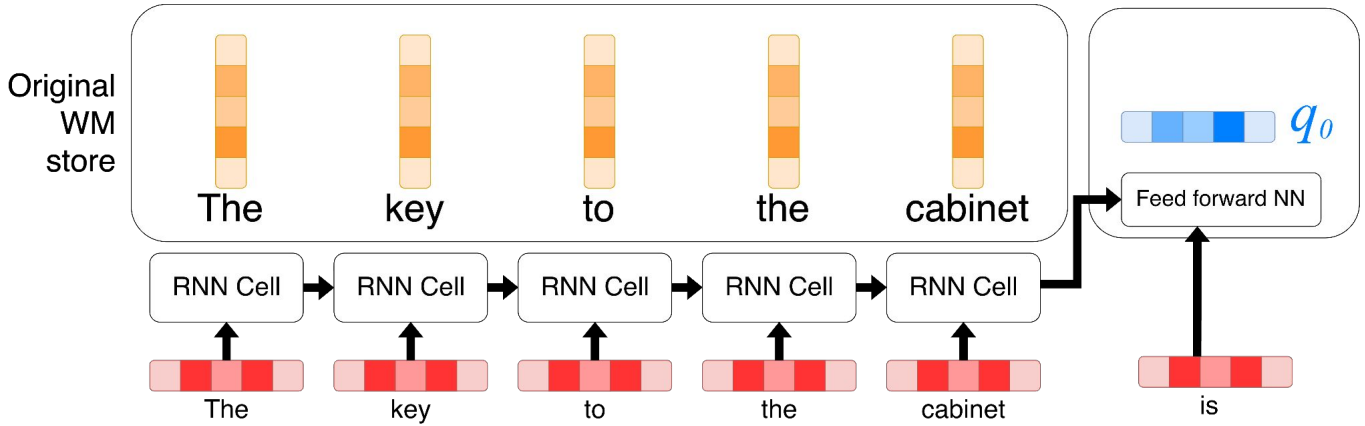
Penn Treebank parsing performance

- Unlabeled attachment score (UAS):
  - Our model : **91%**
  - spacy dependency parser (en-web-trf): **95%**
- Our parser is pretty good!

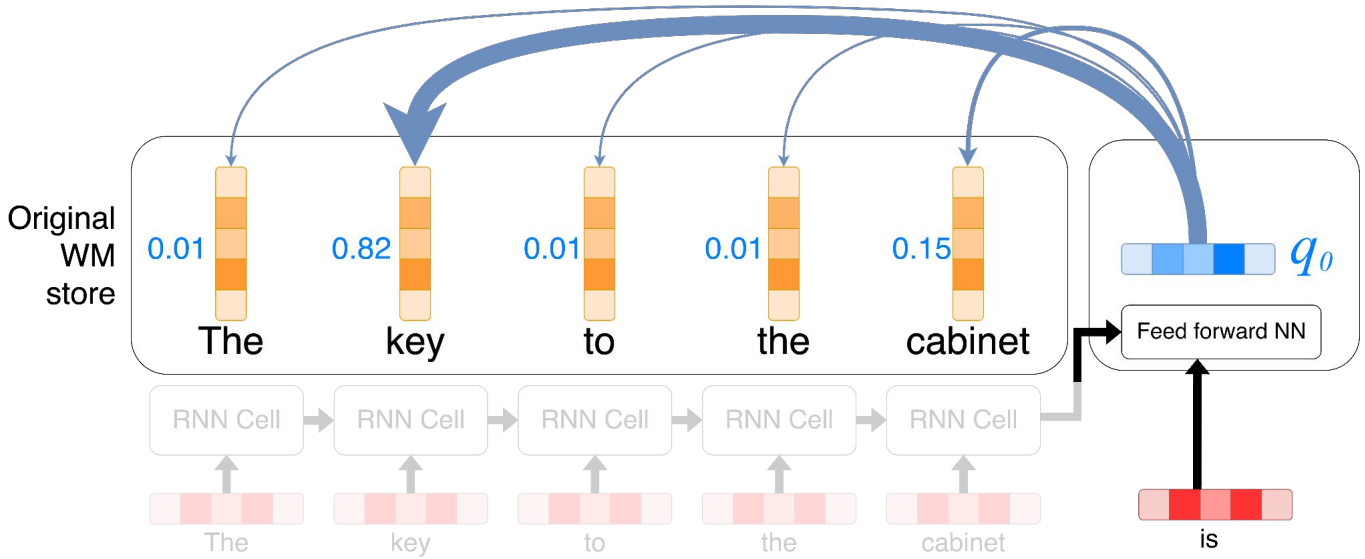
Experiment 1: What information does the model use to resolve simple subject-verb dependencies?

- We use activation patching to uncover the features the model uses to retrieve subjects

Experiment 1: What information does the model use to resolve simple subject-verb dependencies?

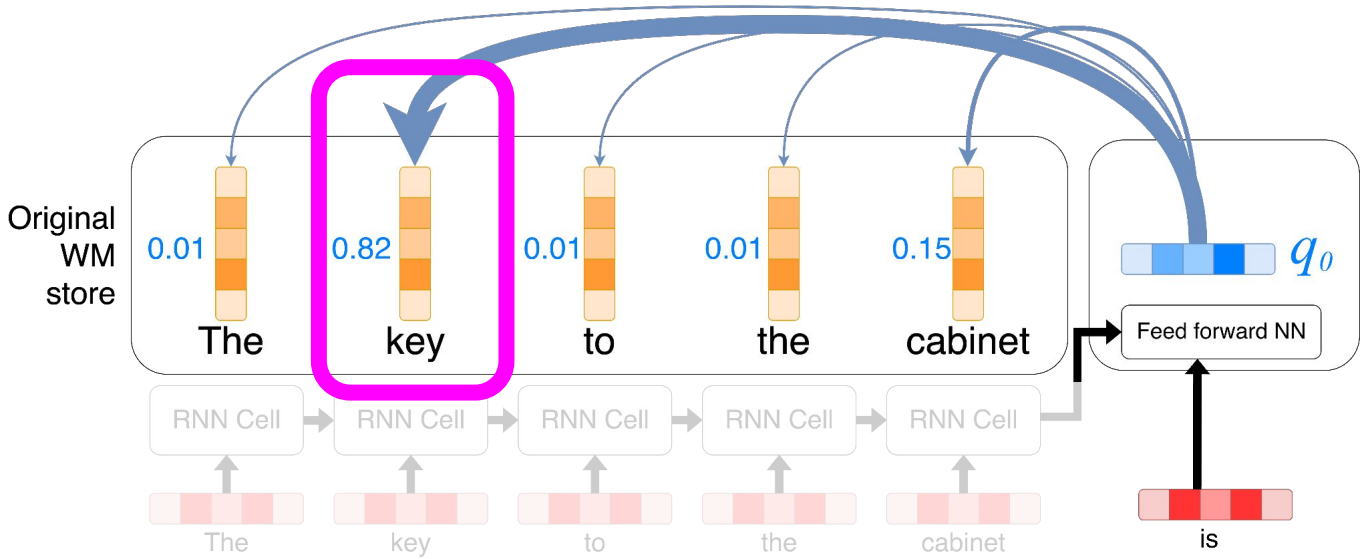


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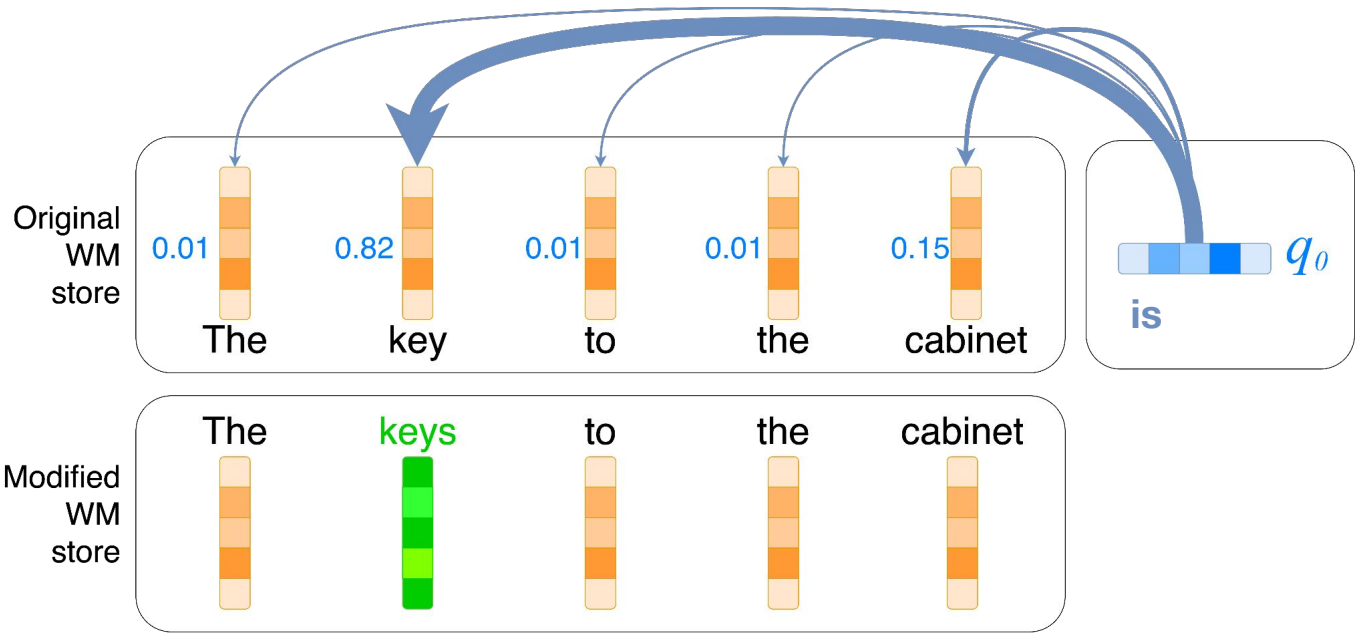
Experiment 1: What information does the model use to resolve simple subject-verb dependencies?

$a(q_0, \text{key}) = 0.82$



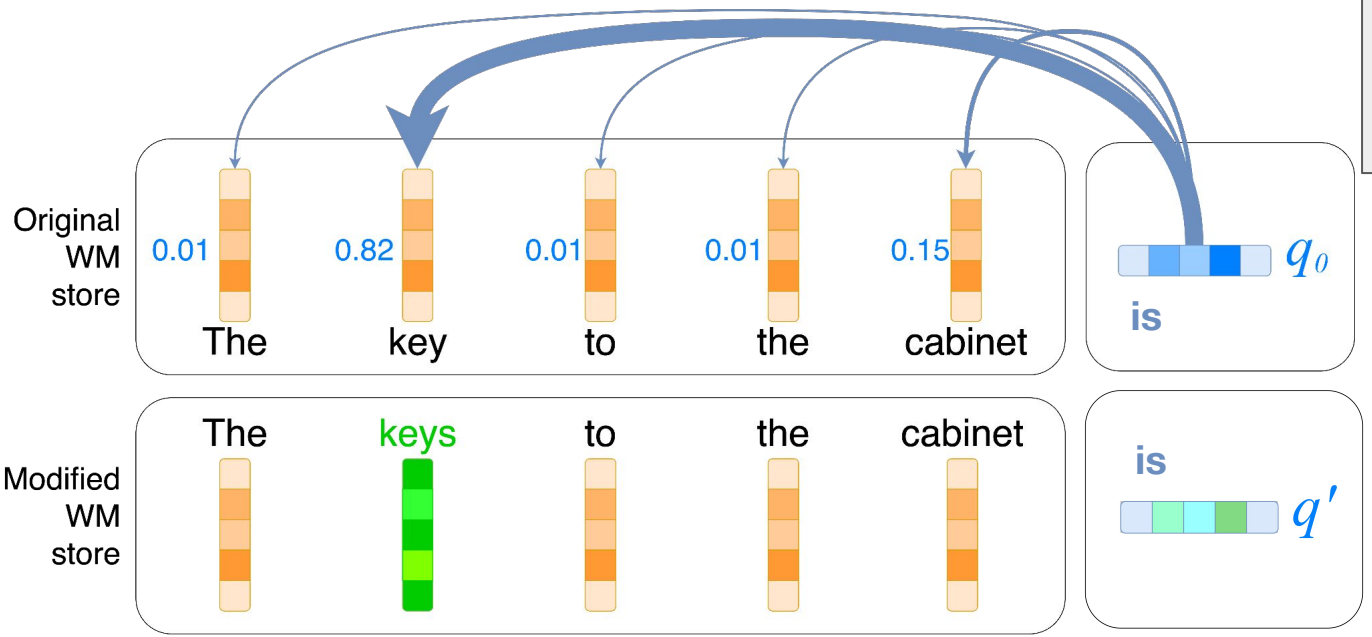
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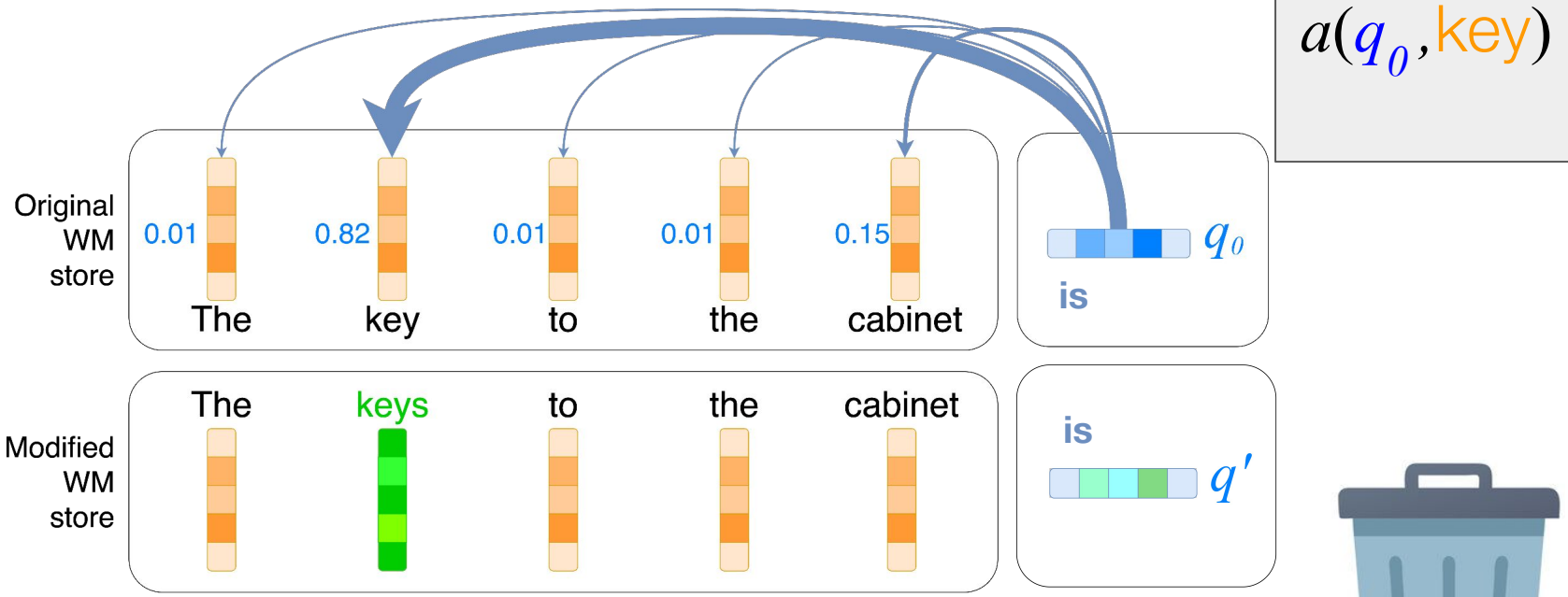
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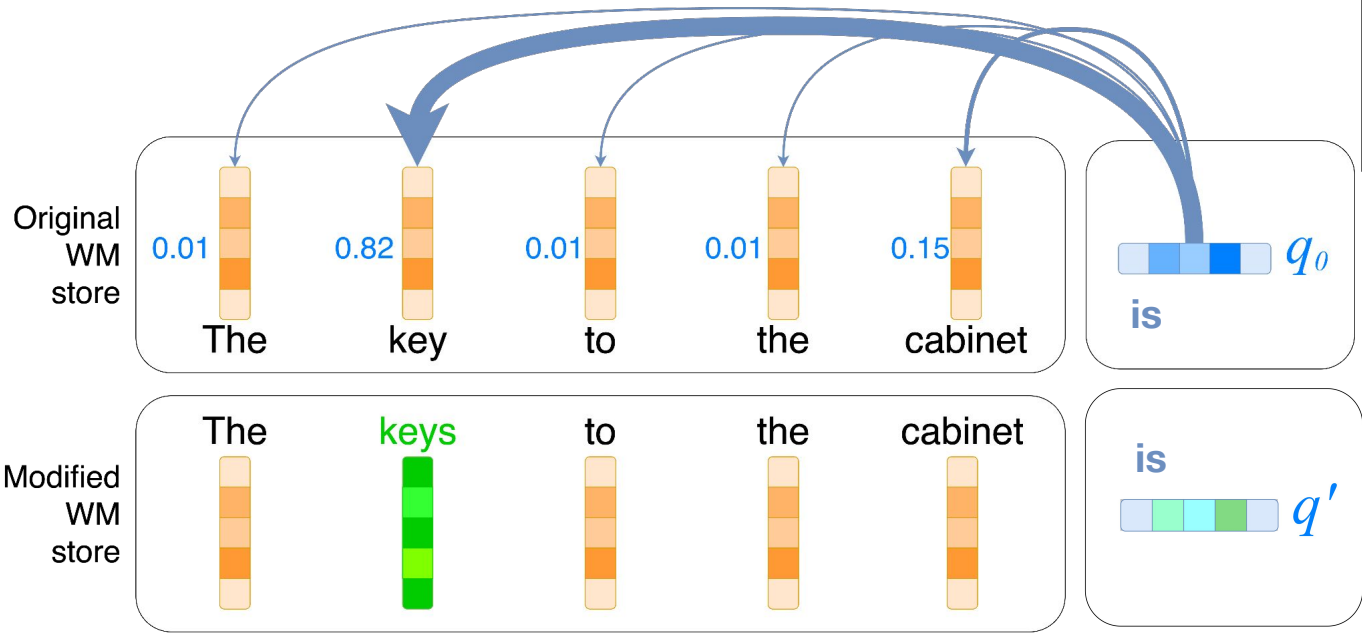
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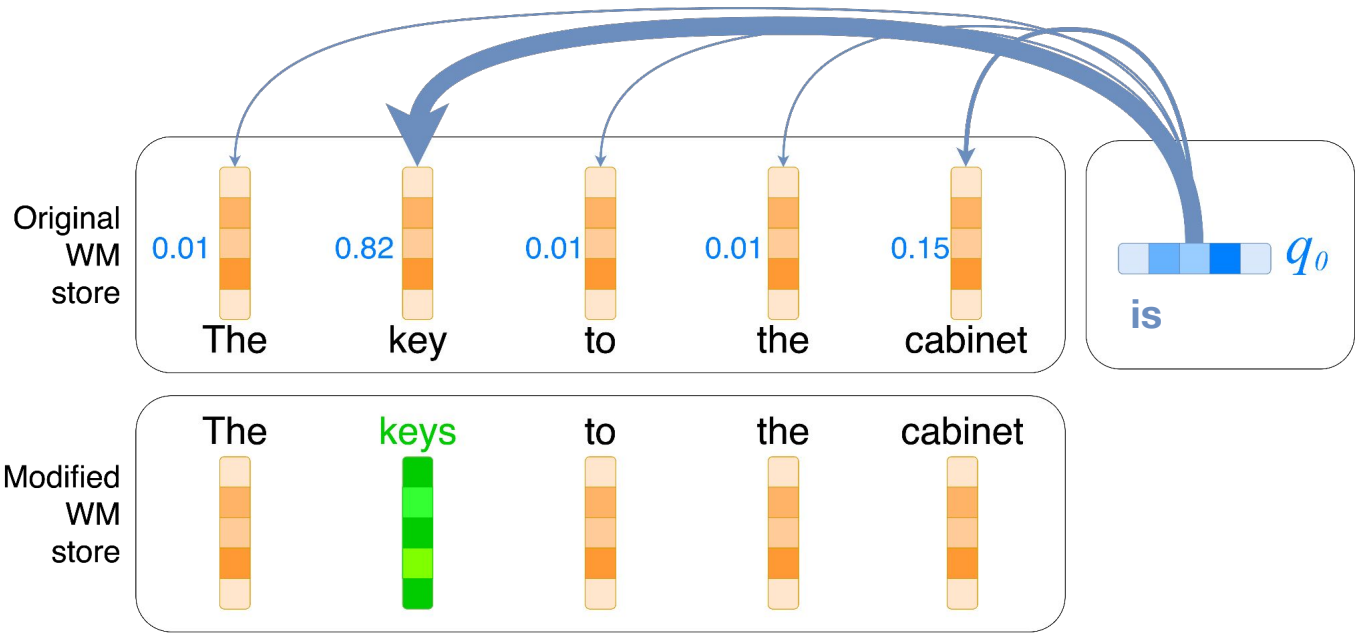
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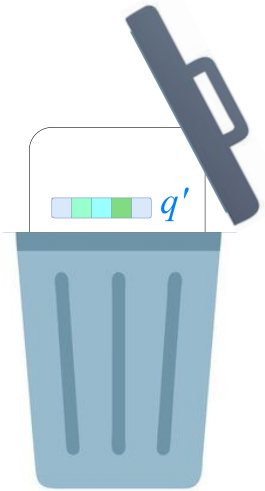




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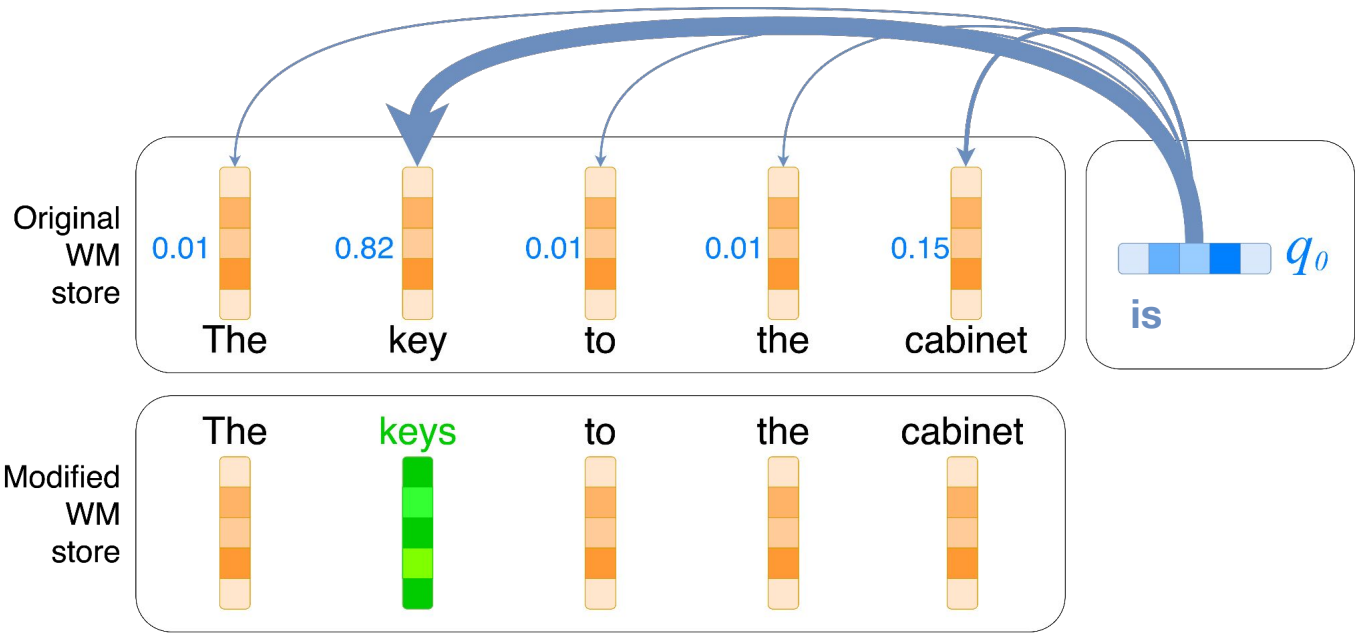
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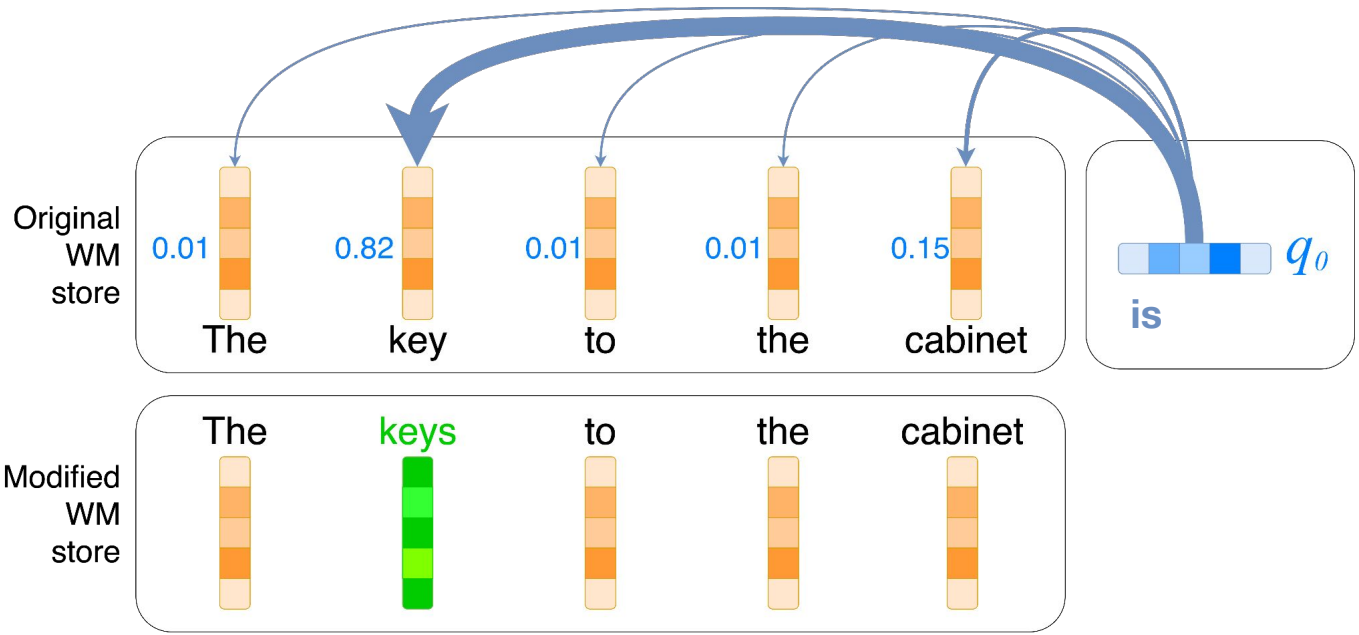
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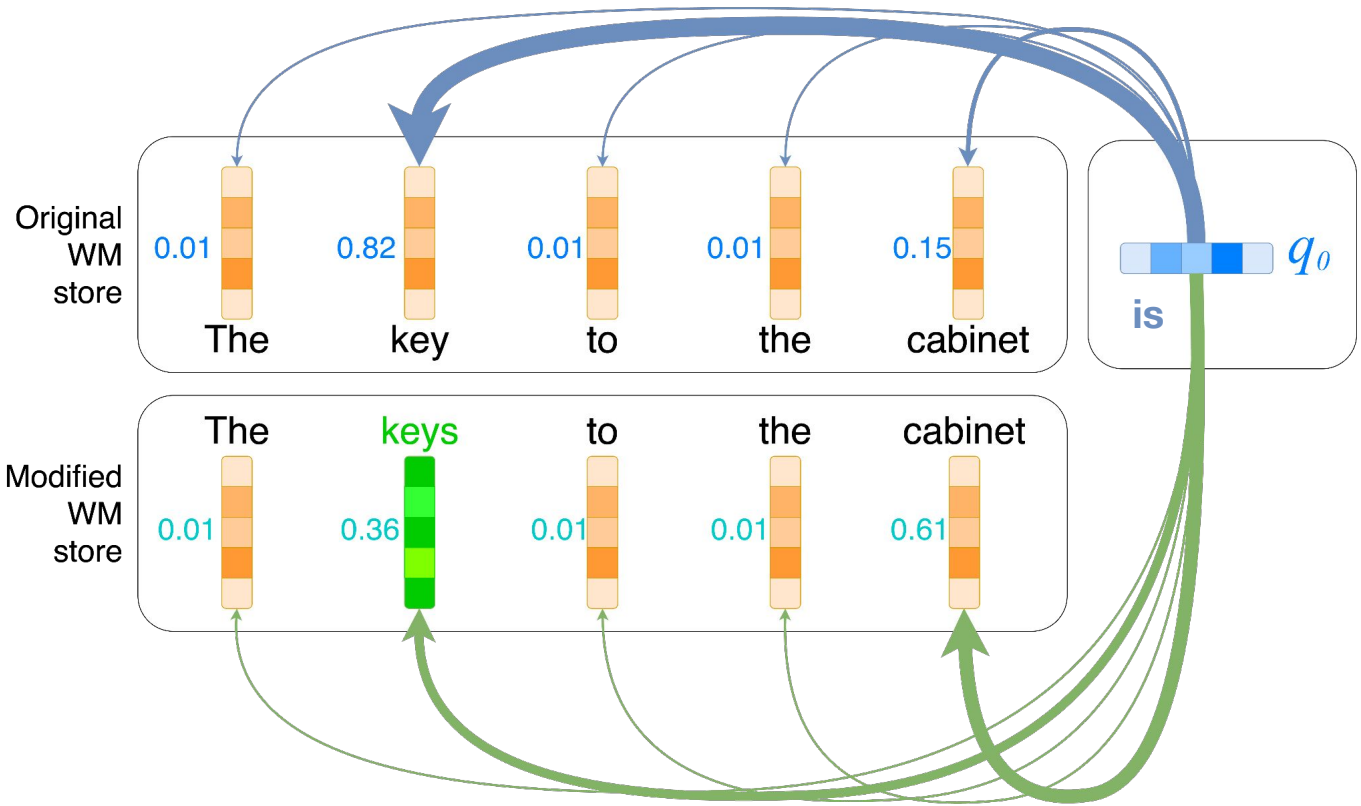
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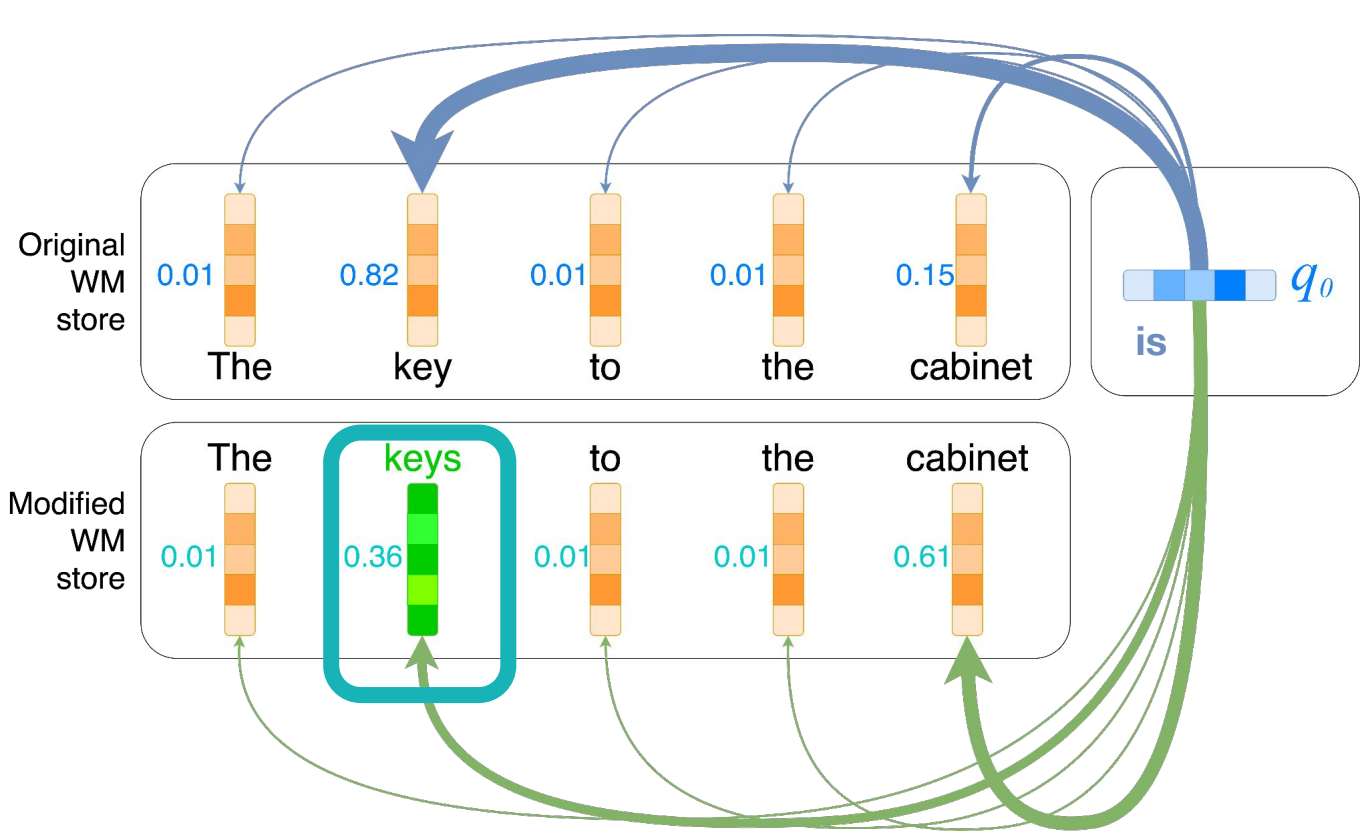


Experiment 1: What information does the model use to resolve simple subject-verb dependencies?

$a(q_0, \text{key}) = 0.82$



Experiment 1: What information does the model use to resolve simple subject-verb dependencies?



$a(q_0, \text{key}) = 0.82$

$a(q_0, \text{keys}) = 0.36$

Experiment 1: What information does the model use to resolve simple subject-verb dependencies?

$a(q_0, \text{key}) = 0.82$
$a(q_0, \text{keys}) = 0.36$

If:

$$a(q_0, \text{key}) >> a(q_0, \text{keys})$$

Then:

Model uses `[+/- plural]` cue

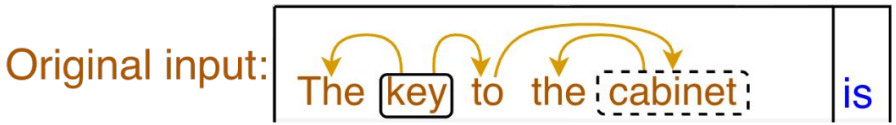
Experiment 1: What information does the model use to resolve simple subject-verb dependencies?

- 24 items From Wagers et al. (2009) of the form:

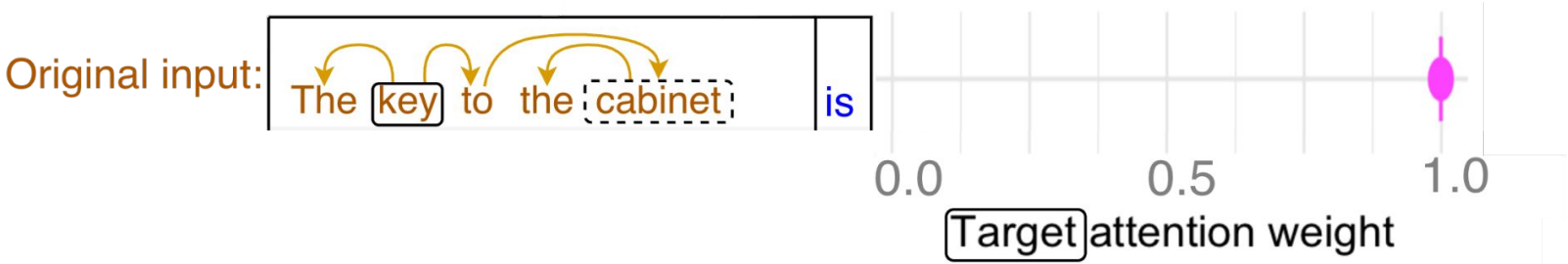
  The N1 P the N2 V

- We report attention to the target word given each intervention

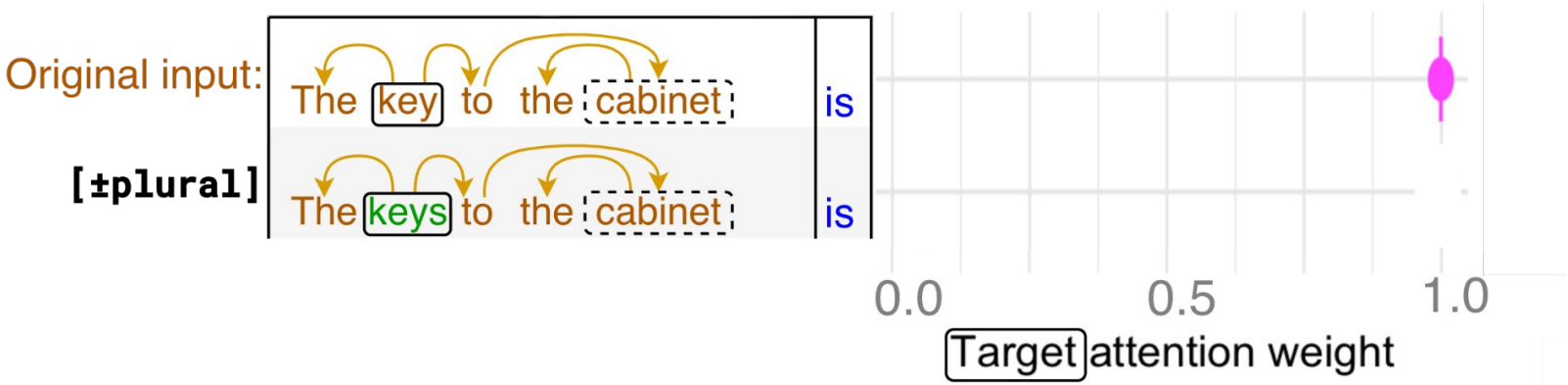
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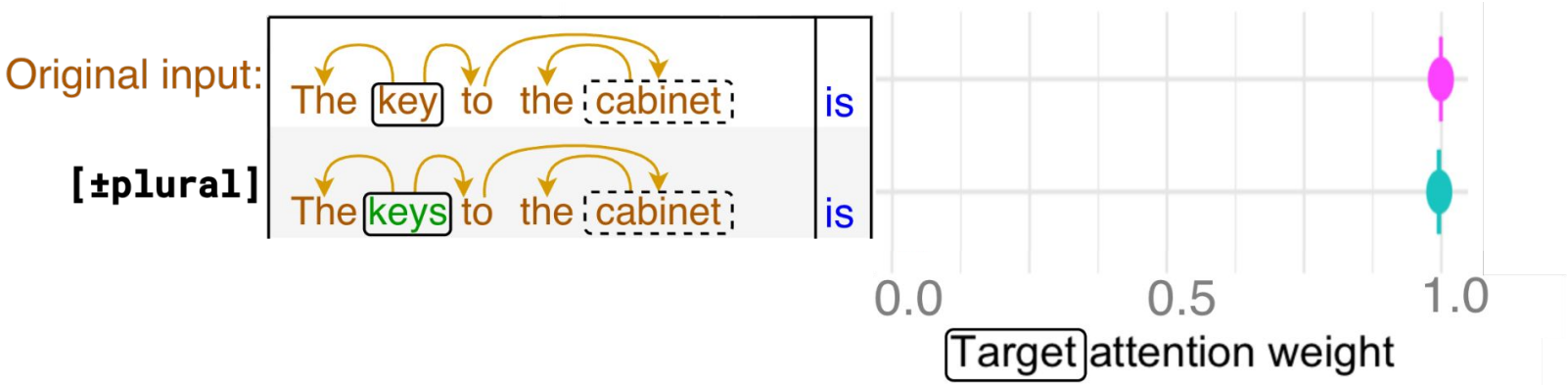
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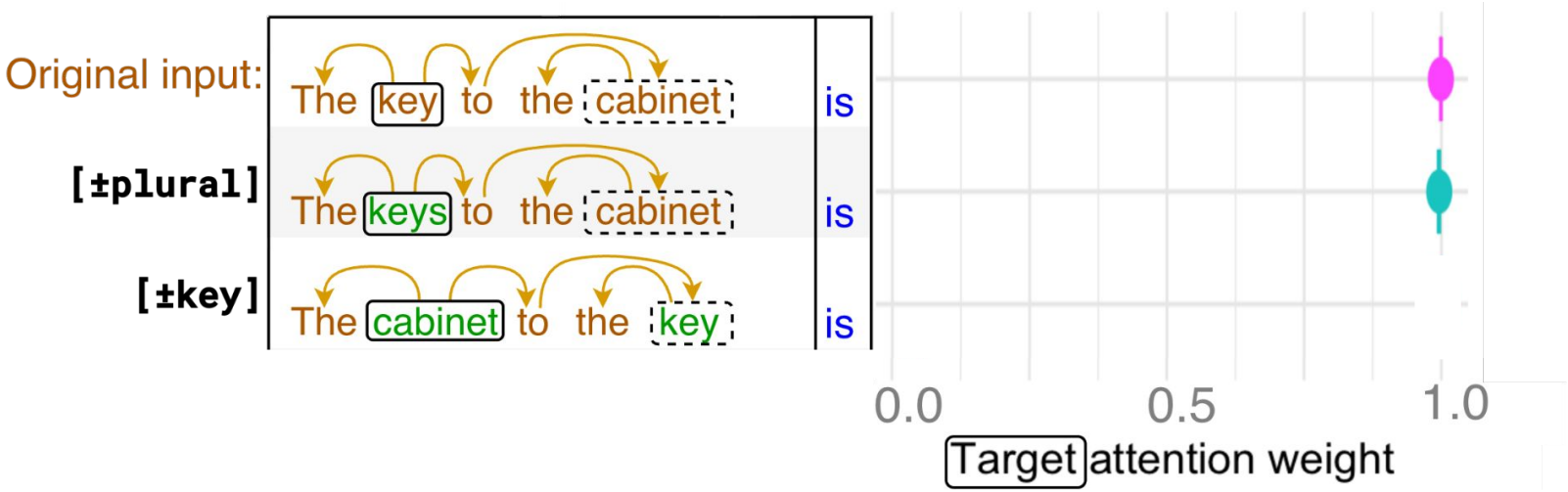
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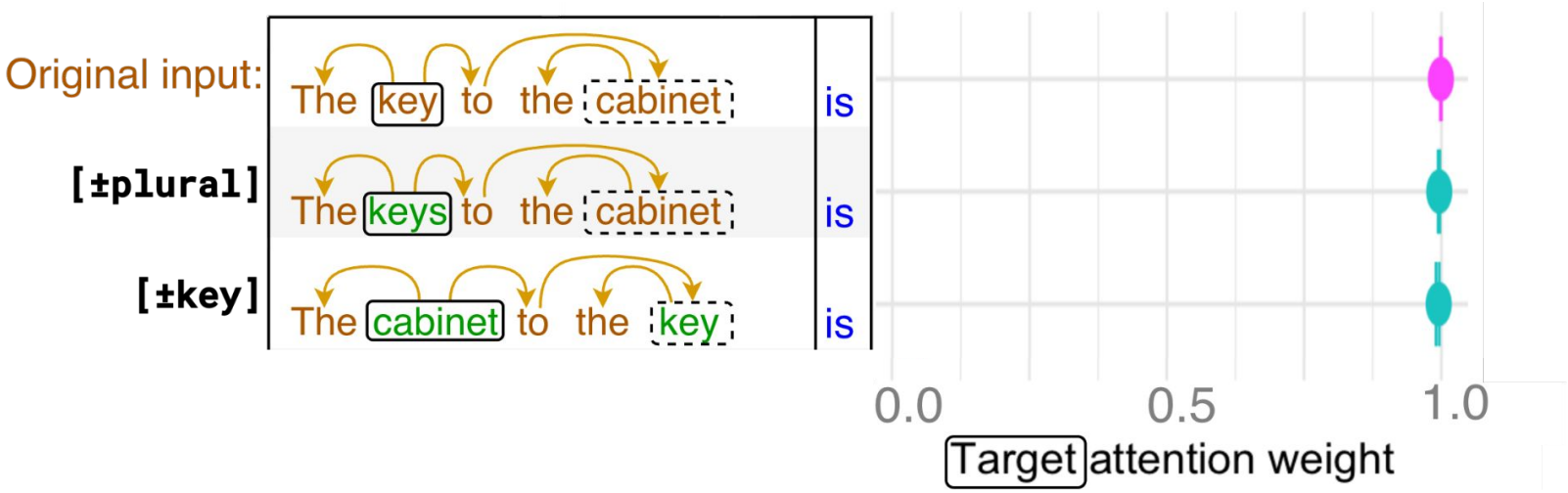
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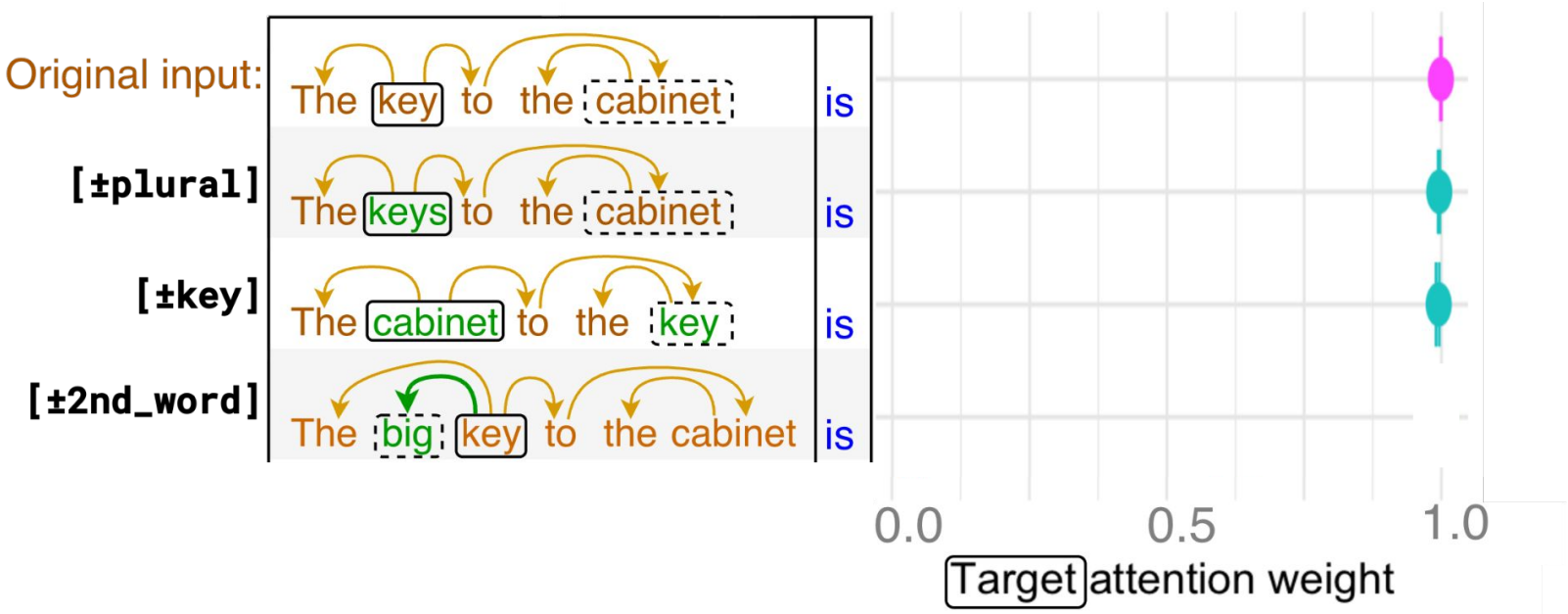
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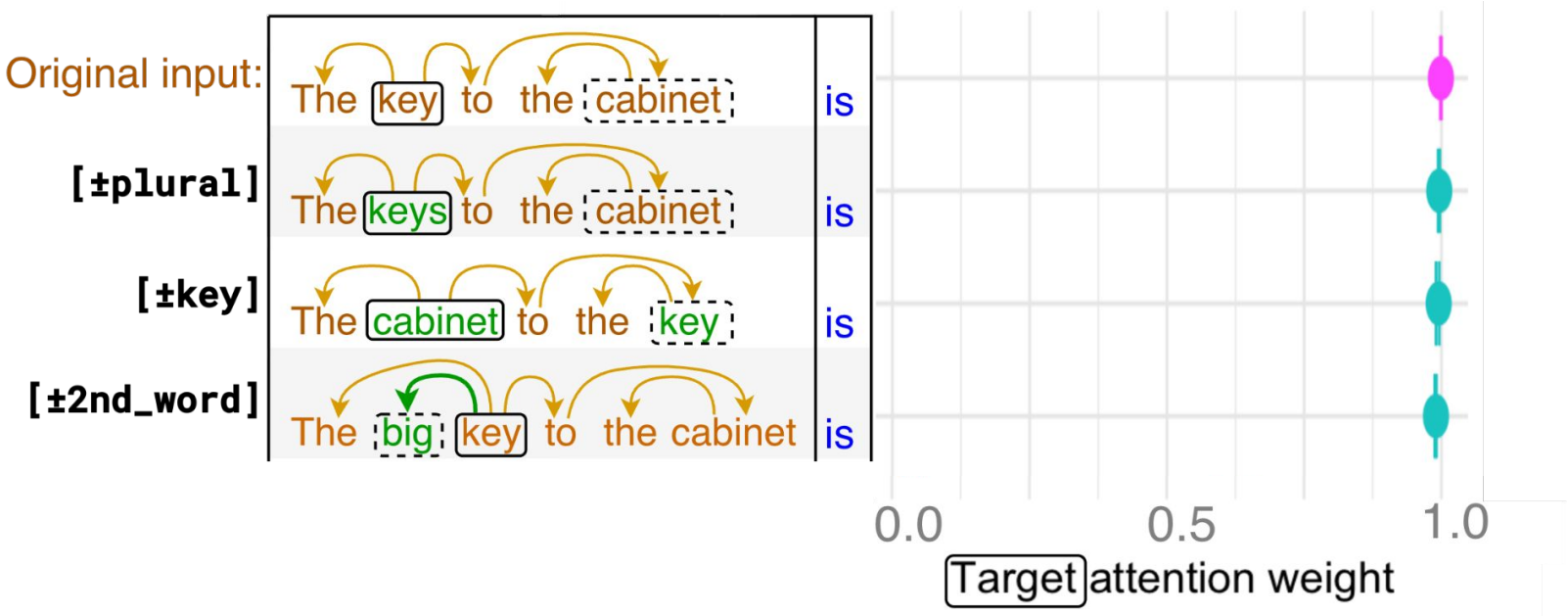
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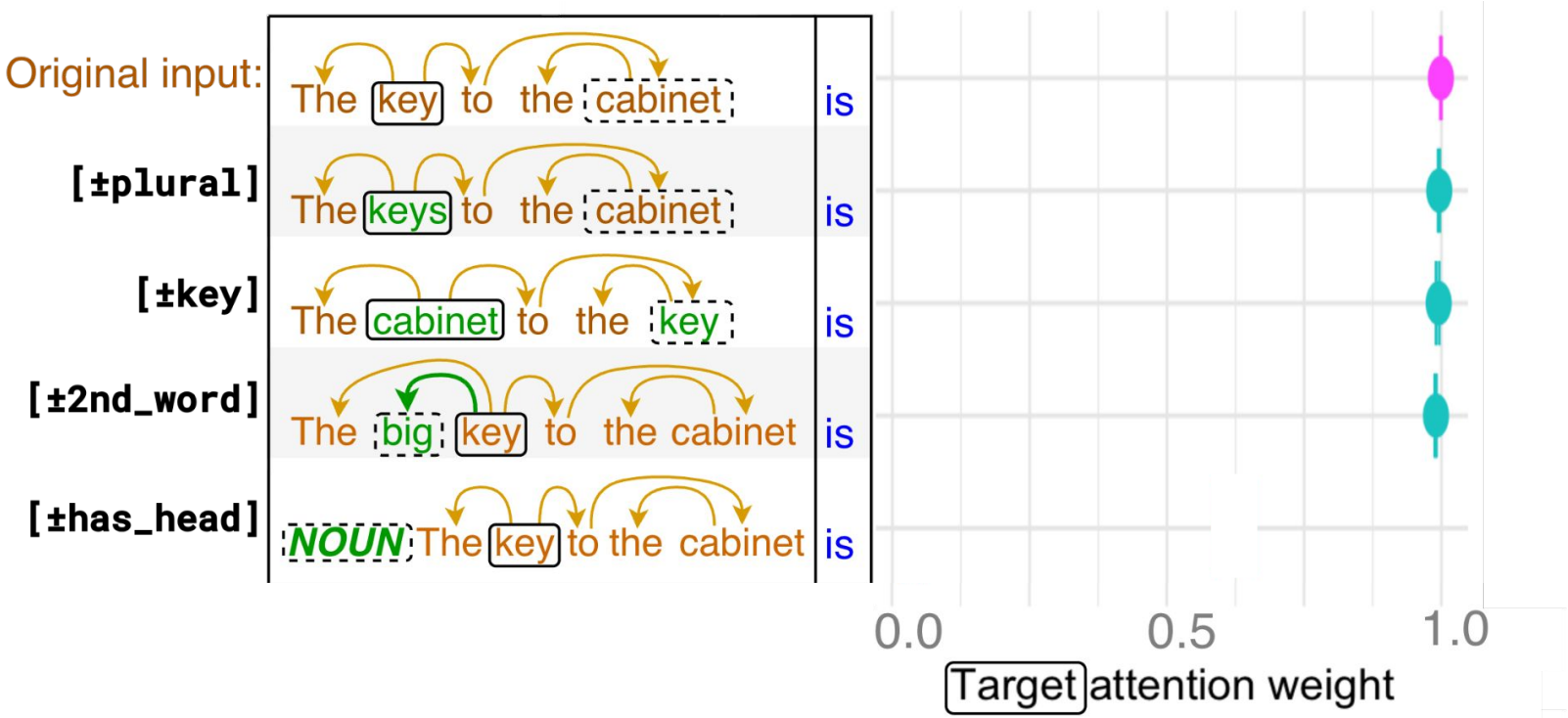
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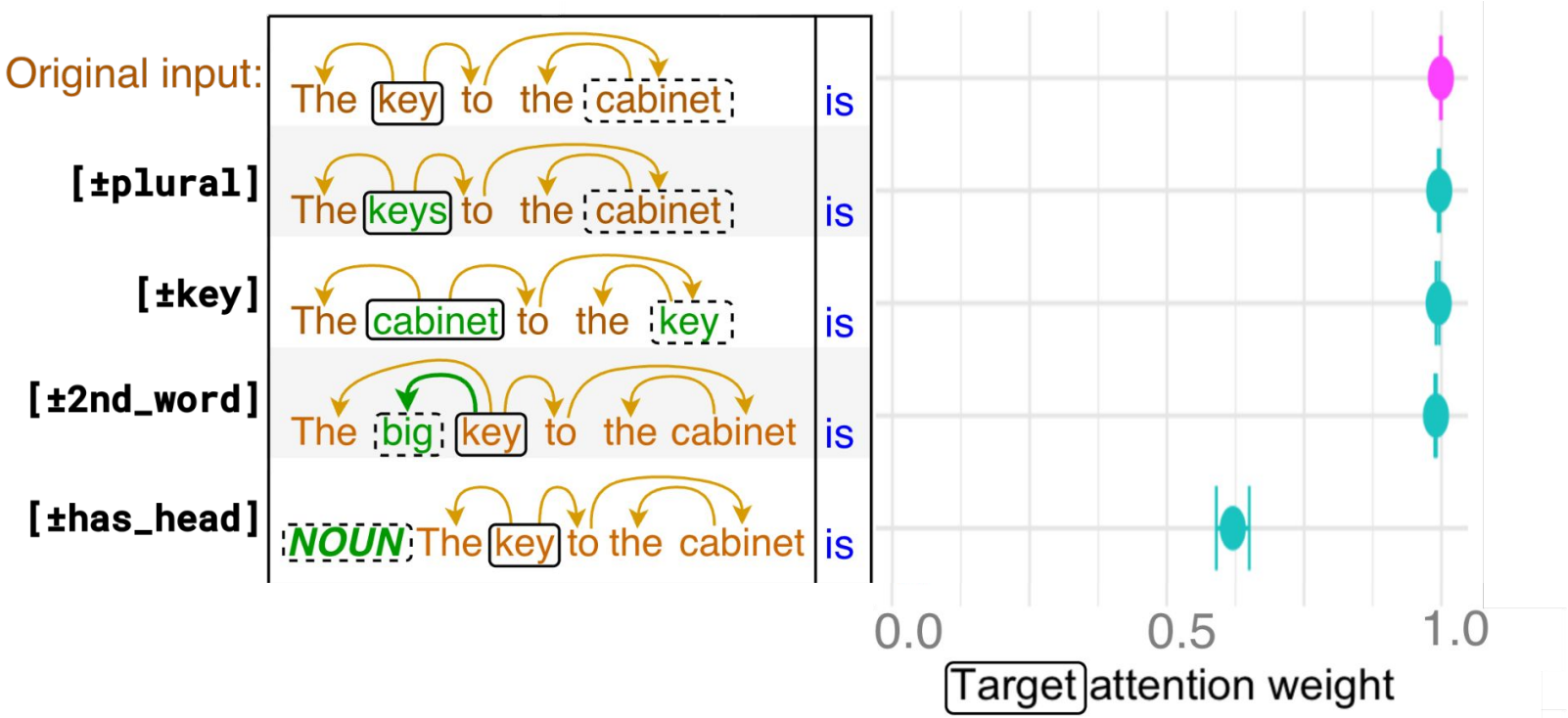
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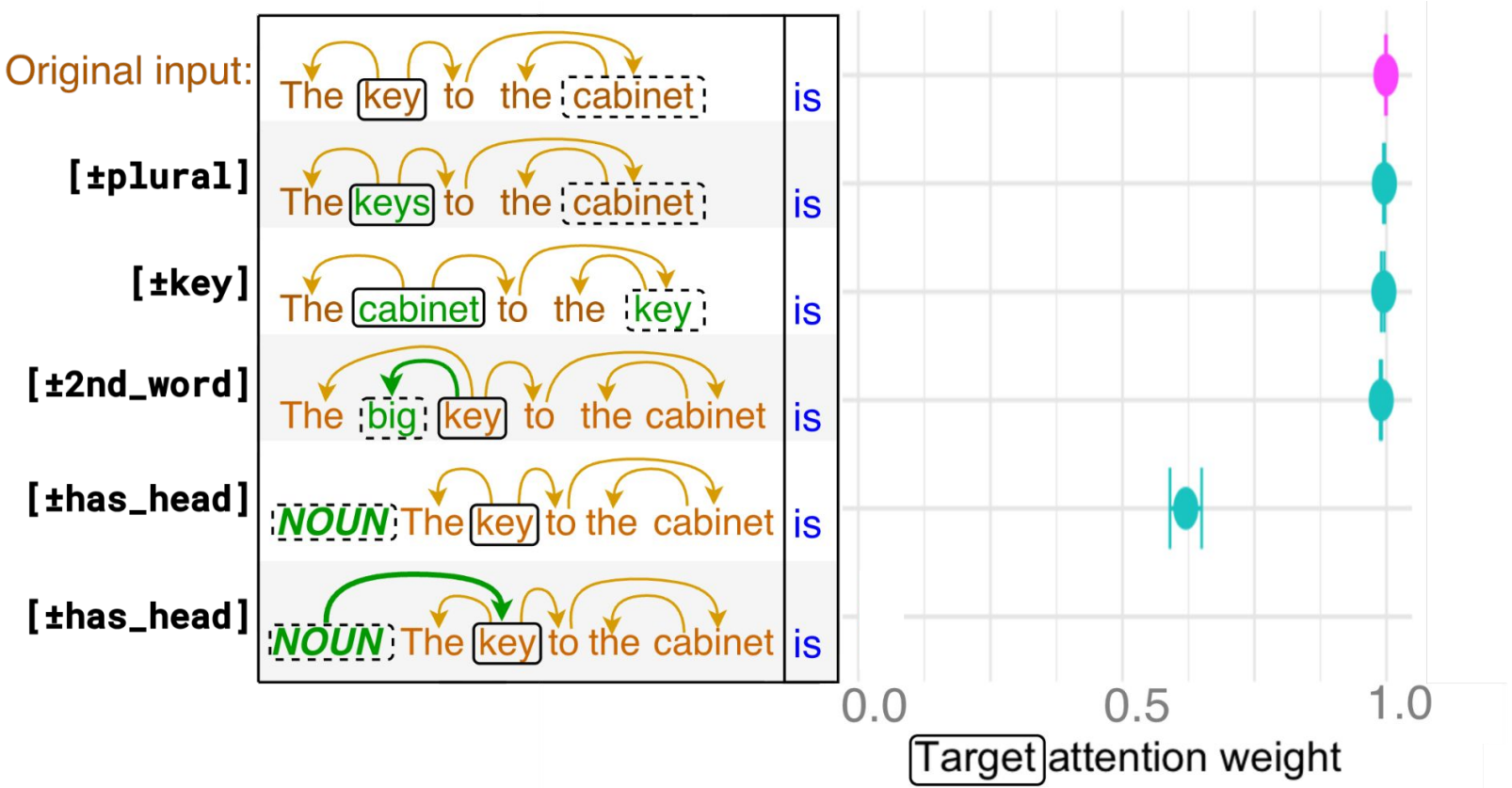
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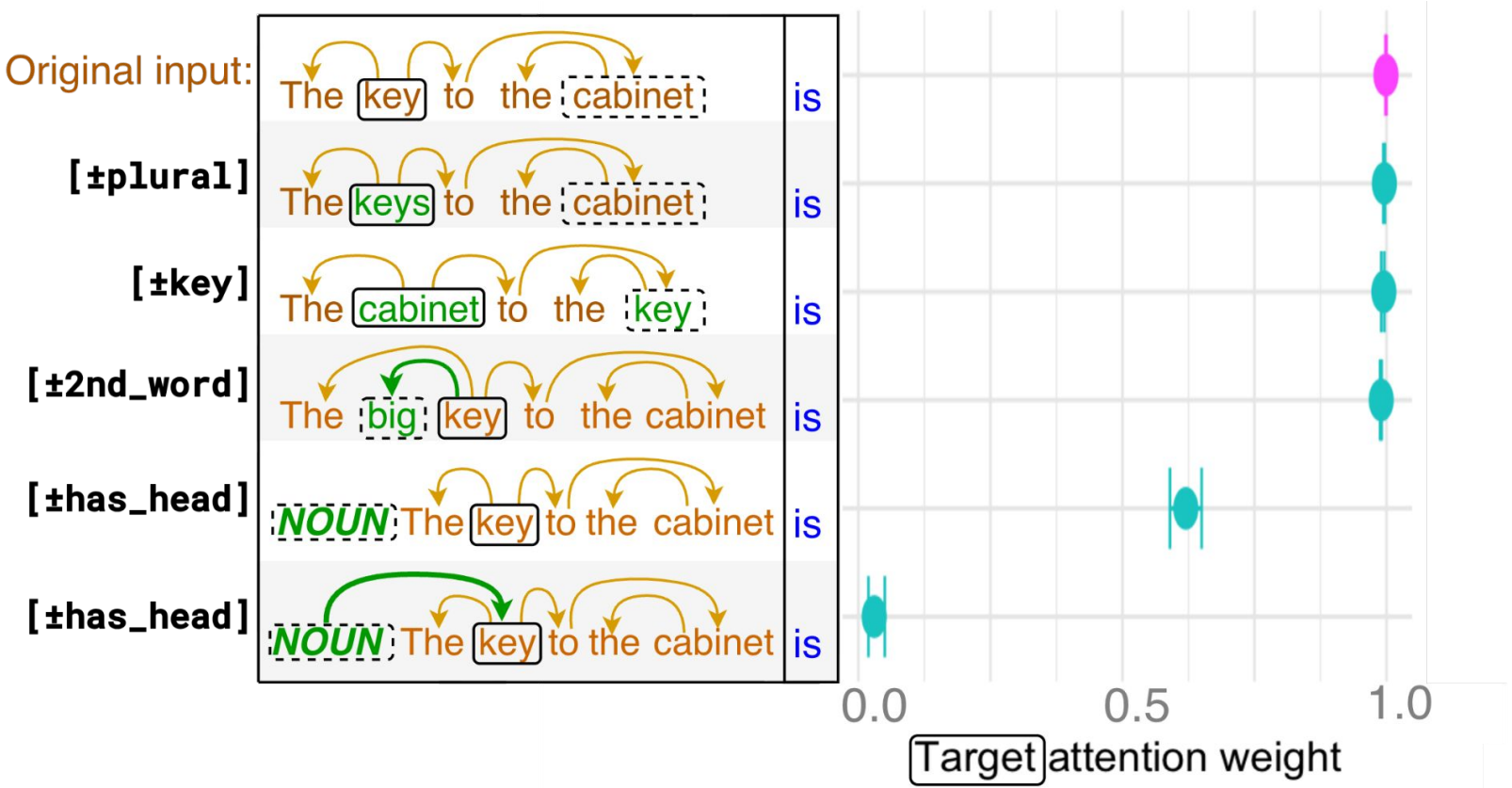
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Interim summary

The model does not use “surface” features at all in simple sentences

➔ Why? Simple to encode, but too many false positives

Instead, it probes for nouns that have not yet been linked with their syntactic head

➔ Why? **`[-has_head]`** is simple to encode AND highly discriminative

Interim summary

The model does not use “surface” features at all in simple sentences

➔ Why? Simple to encode, but too many false positives

Instead, it probes for nouns that have not yet been linked with their syntactic head

➔ Why? **`[-has_head]`** is simple to encode AND highly discriminative

Feature	Precision
`[±plural]`	44.6%
`[+subject]`	85.6%
`[-has_head]`	99.2%

Prediction:

- Interference should only arise when there are **multiple unresolved** long-distance dependencies (multiple center embedding)
- These are rare in training data, and notoriously difficult for humans

- Do models ever make use of other features when **`[±has_head]`** isn't enough?

Hahn et al. (2022)

Multiple center embedding

Single embedding (Depth = 2):

> The fact that [the doctor annoyed the patient] was surprising.

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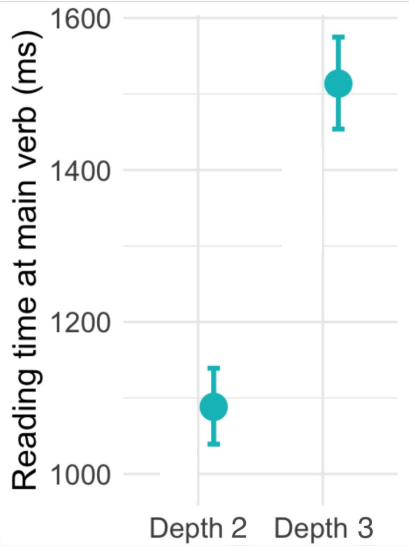
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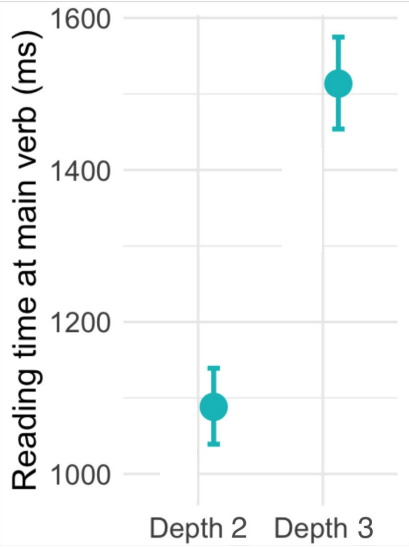
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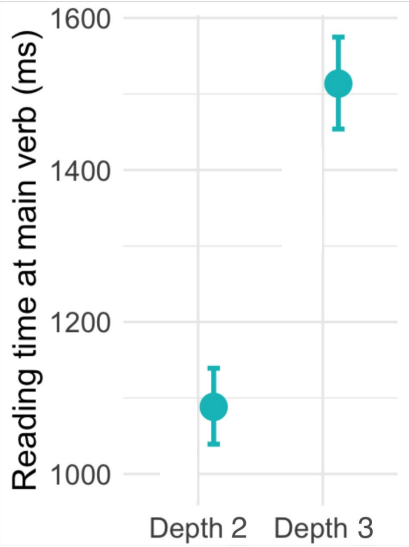
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Local dependency; no retrieval needed (Häussler & Bader 2015)



Hahn et al. (2022)

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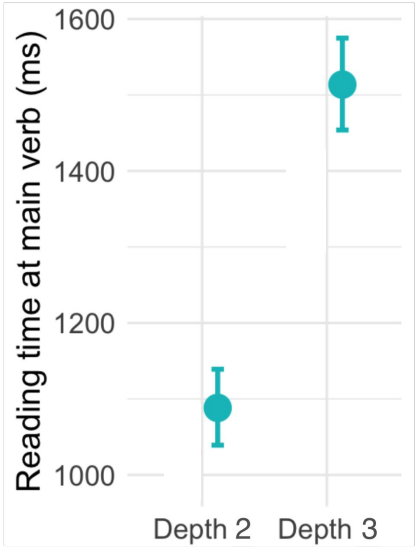
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Double embedding (Depth = 3):

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???



Long-distance; retrieval needed; but multiple `[-has_head]` items!

Hahn et al. (2022)

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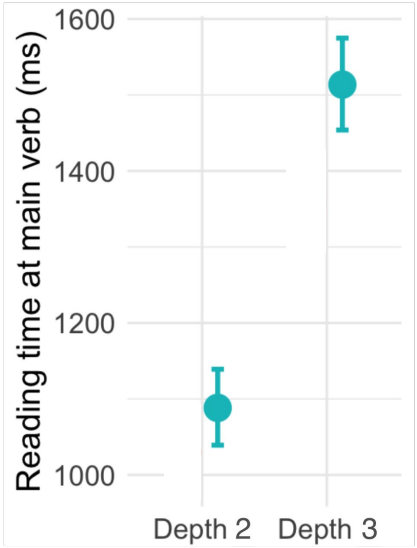
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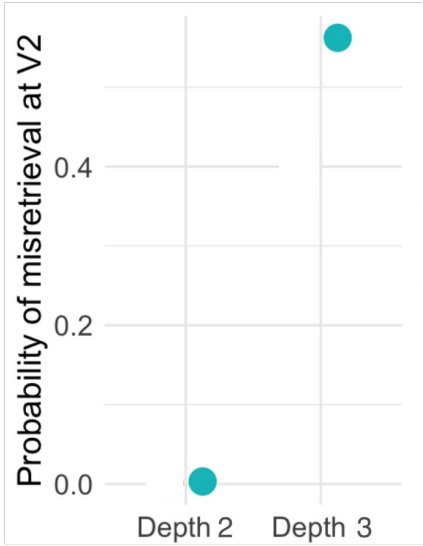
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???





Hahn et al. (2022)

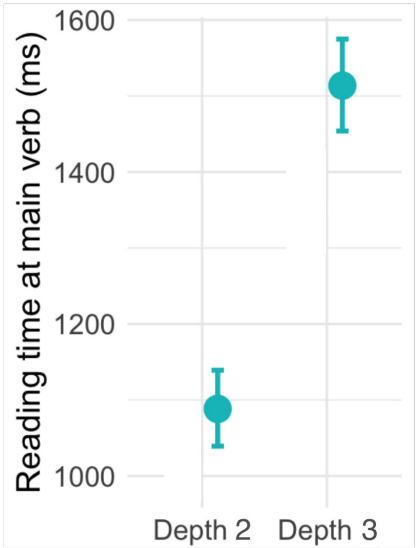
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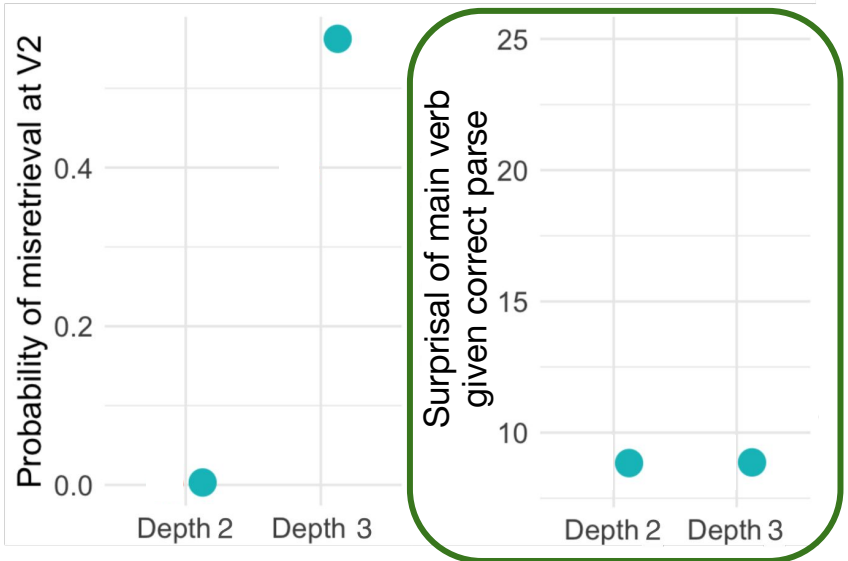
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Hahn et al. (2022)

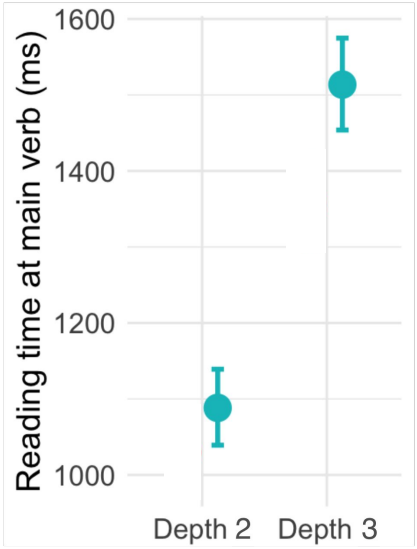
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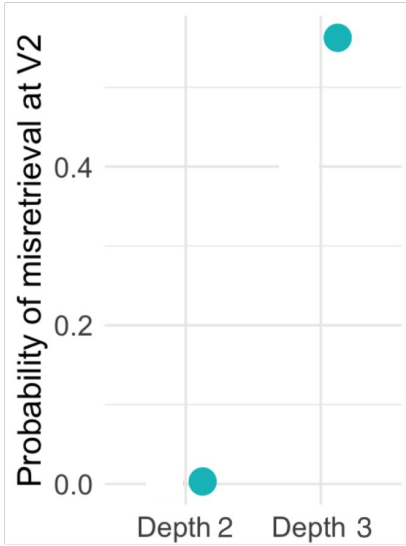
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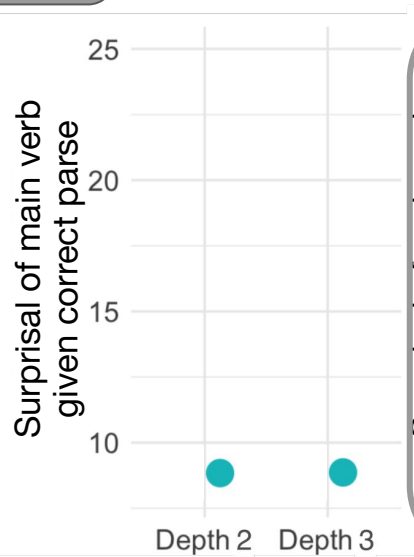
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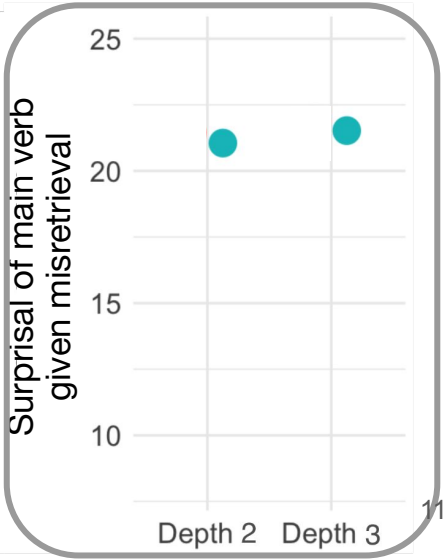
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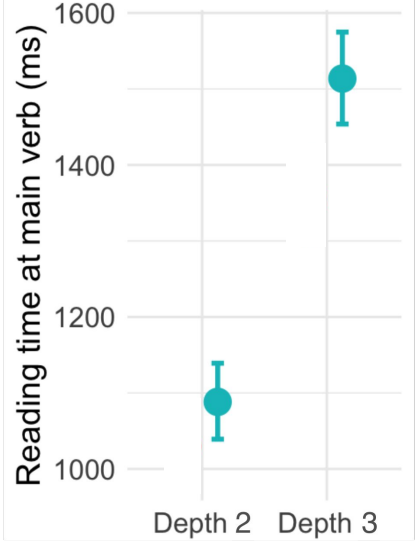
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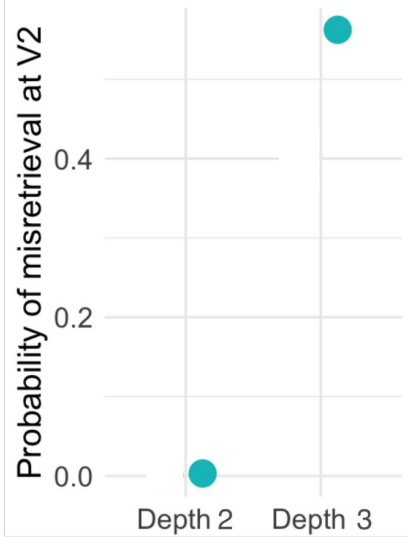
Single embedding (Depth = 2):

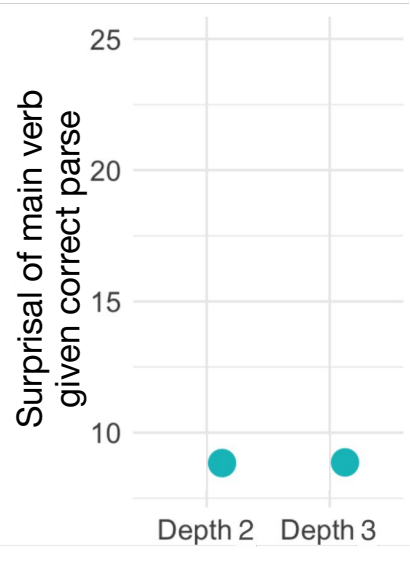
The fact that [the doctor {cured / annoyed} the patient] was surprising.

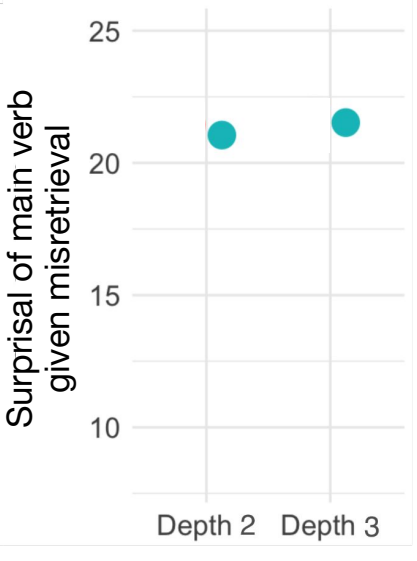
Double embedding (Depth = 3):

The fact that [the doctor who [the lawyer distrusted] {cured / annoyed} the patient] was surprising.









Hahn et al. (2022)

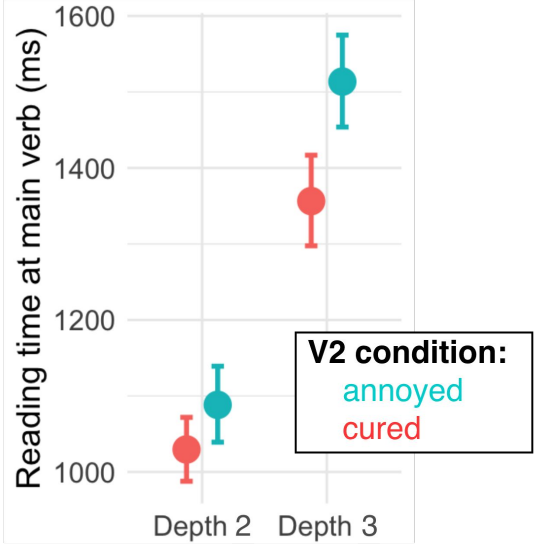
Multiple center embedding

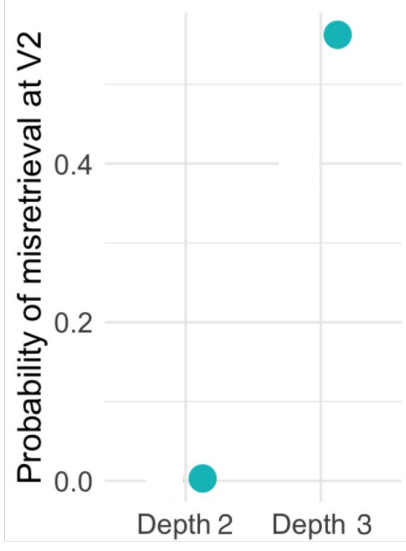
Single embedding (Depth = 2):

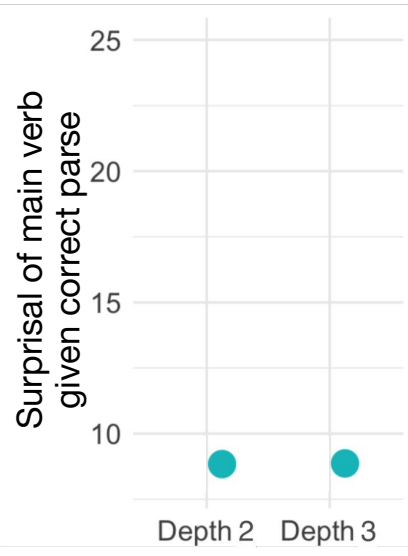
The fact that [the doctor {**cured** / **annoyed**} the patient] **was** surprising.

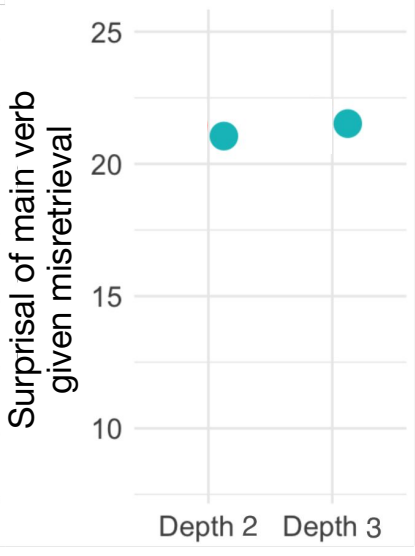
Double embedding (Depth = 3):

The fact that [the doctor who [the lawyer distrusted] {**cured** / **annoyed**} the patient] **was** surprising.









Hahn et al. (2022)

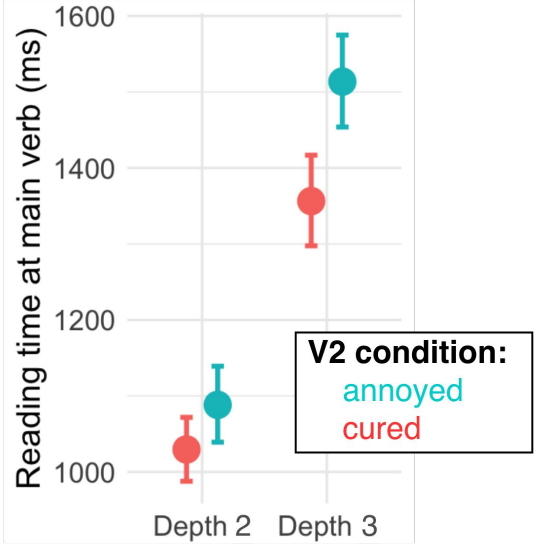
Multiple center embedding

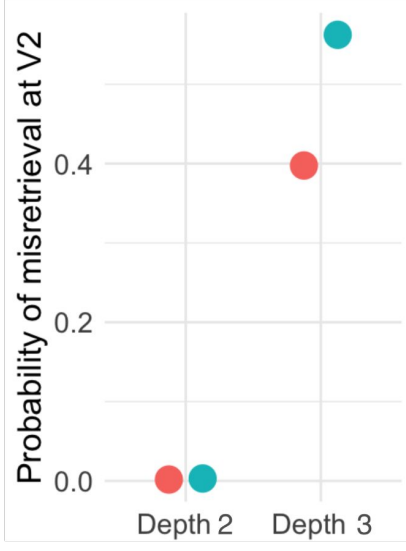
Single embedding (Depth = 2):

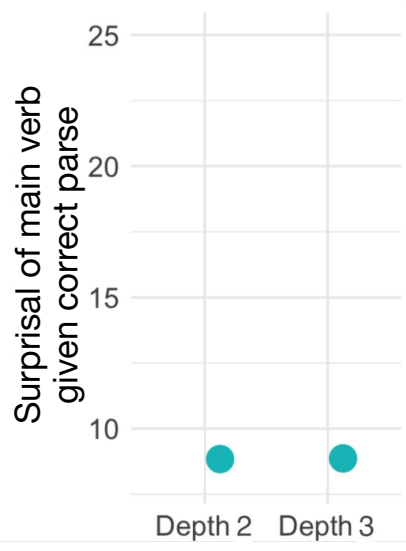
The fact that [the doctor {**cured** / **annoyed**} the patient] was surprising.

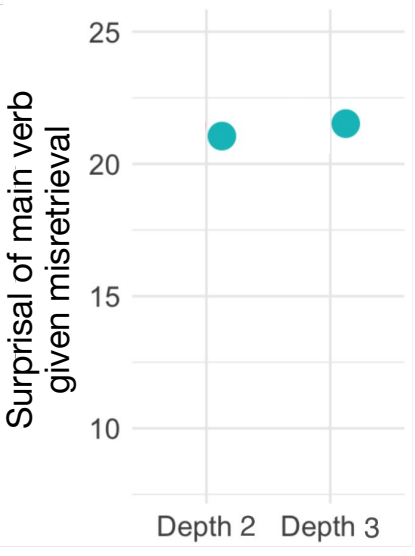
Double embedding (Depth = 3):

The fact that [the doctor who [the lawyer distrusted] {**cured** / **annoyed**} the patient] was surprising.









Hahn et al. (2022)

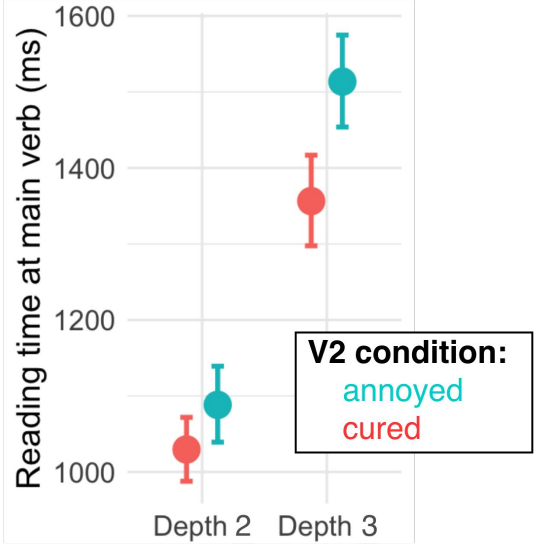
Multiple center embedding

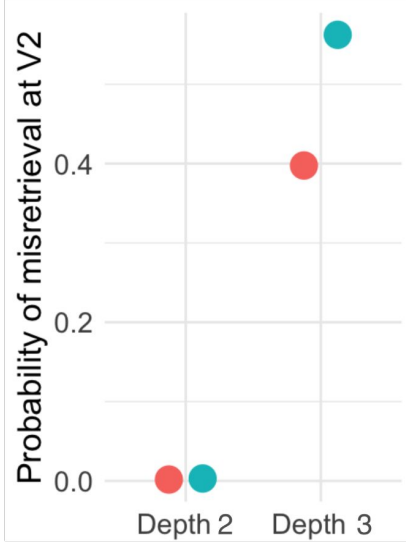
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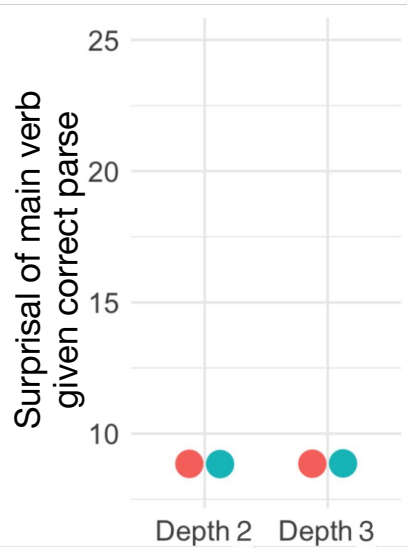
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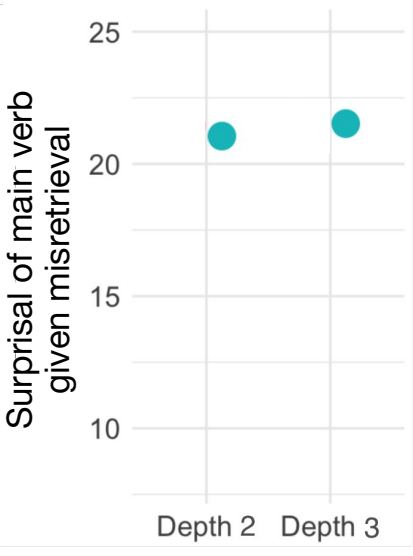
Double embedding (Depth = 3):

The fact that [the doctor who [the lawyer distrusted] {**cured** / **annoyed**} the patient] **was** surprising.









Hahn et al. (2022)

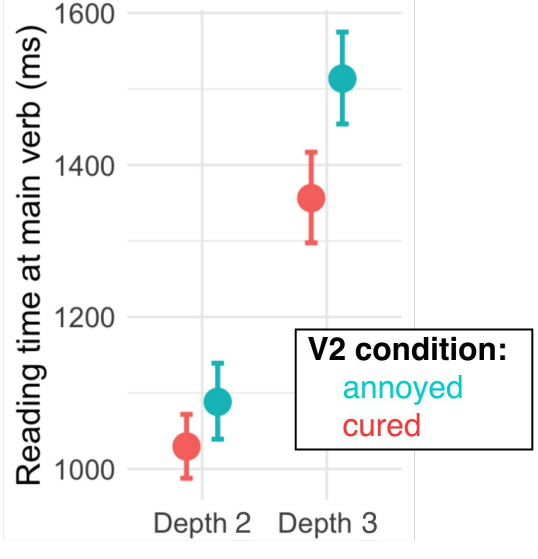
Multiple center embedding

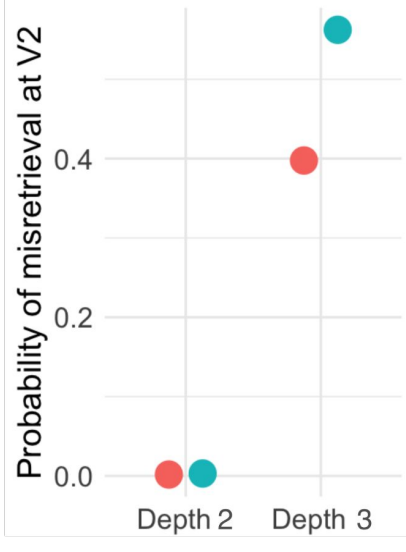
Single embedding (Depth = 2):

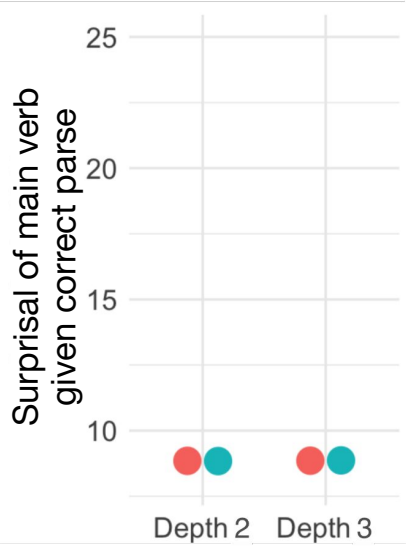
The fact that [the doctor {**cured** / **annoyed**} the patient] **was** surprising.

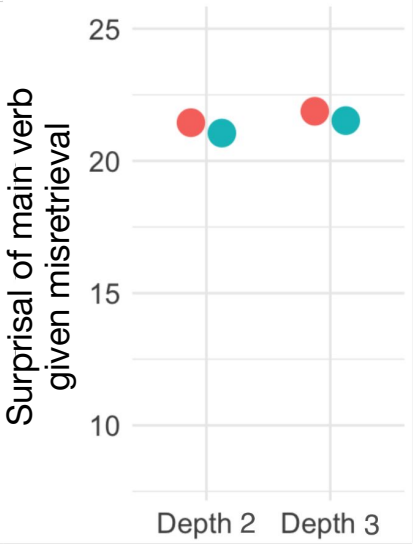
Double embedding (Depth = 3):

The fact that [the doctor who [the lawyer distrusted] {**cured** / **annoyed**} the patient] **was** surprising.









Conclusion

- We can use **neural network** models of dependency resolution to **generate** and **test** hypotheses about memory representations in **humans**
- A rational comprehender prioritizes **highly discriminative** features
- Our model is **data-driven**, but its mechanisms are **interpretable**

Future:

- More direct manipulation of `[±has_head]` feature
- Typological variation (e.g. center embedding in German vs. English)
- Other dependency types (filler-gap, reflexive-antecedent)

Thank you



Brian Dillon



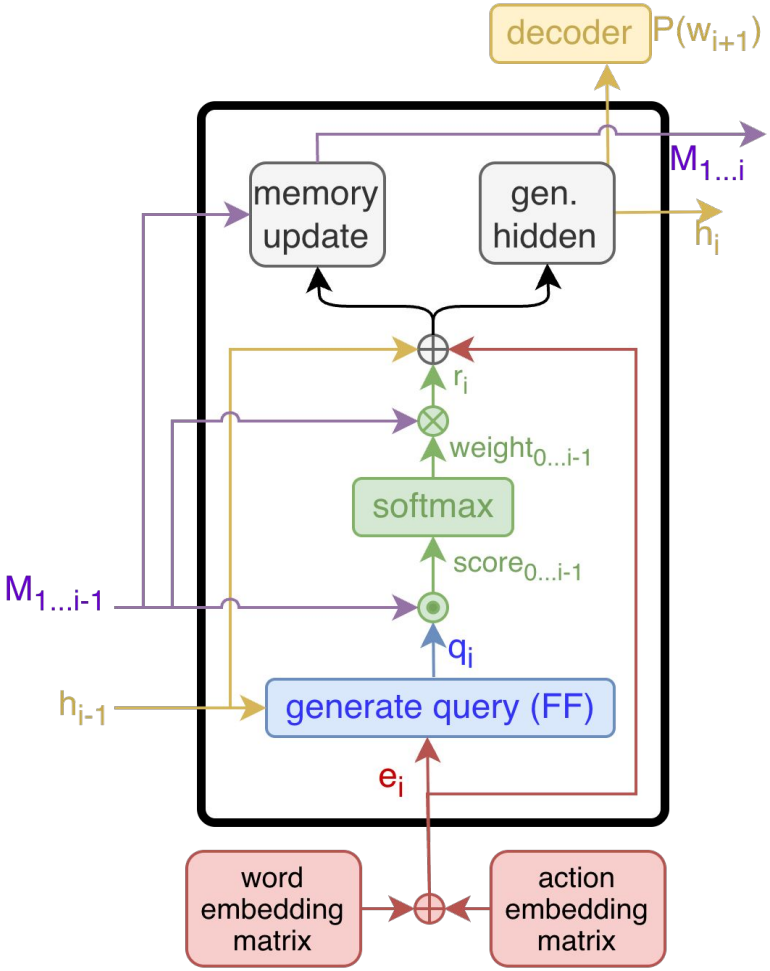
Tal Linzen

NYU IT High Performance Computing

National Science Foundation

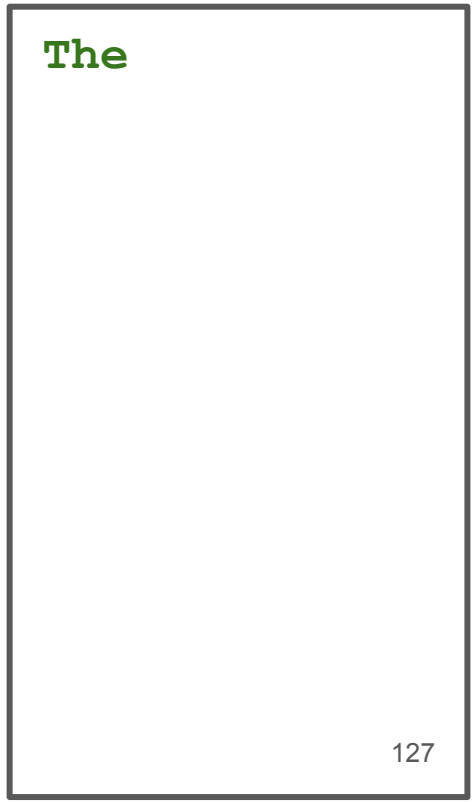
Appendix

Model architecture



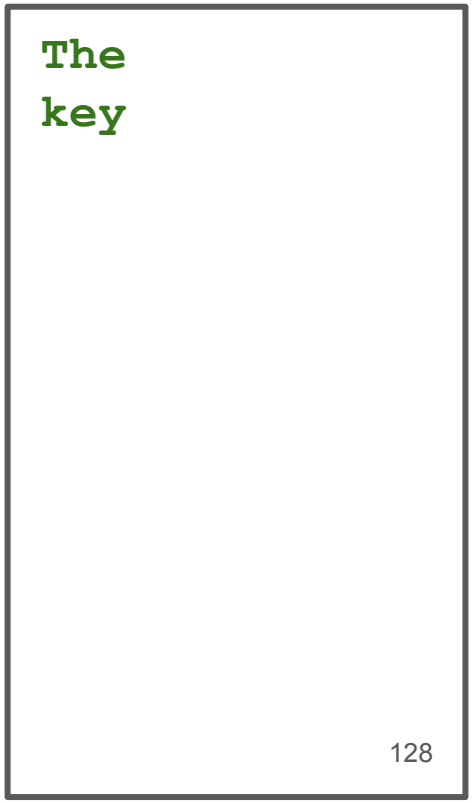
Transition-based dependency parsing (e.g. Nivre 2006)

The

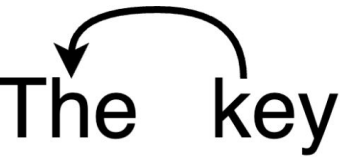


Transition-based dependency parsing (e.g. Nivre 2006)

The key

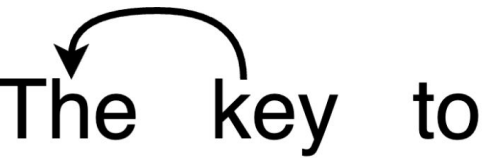


Transition-based dependency parsing (e.g. Nivre 2006)



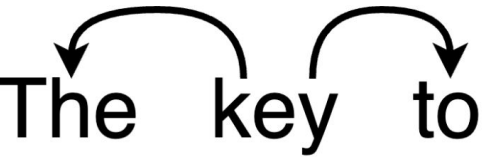
The
key
LA(key,the)

Transition-based dependency parsing (e.g. Nivre 2006)



The
key
LA(key,the)
to

Transition-based dependency parsing (e.g. Nivre 2006)




```
The
key
LA(key,the)
to
RA(key,to)
```

Transition-based dependency parsing (e.g. Nivre 2006)




```
The
key
LA(key,the)
to
RA(key,to)
the
```

Transition-based dependency parsing (e.g. Nivre 2006)



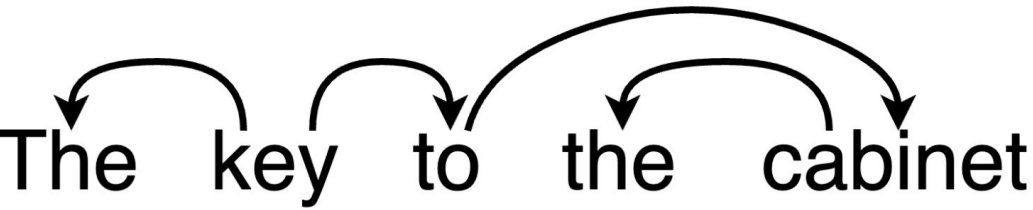

```
The
key
LA(key,the)
to
RA(key,to)
the
cabinet
```

Transition-based dependency parsing (e.g. Nivre 2006)



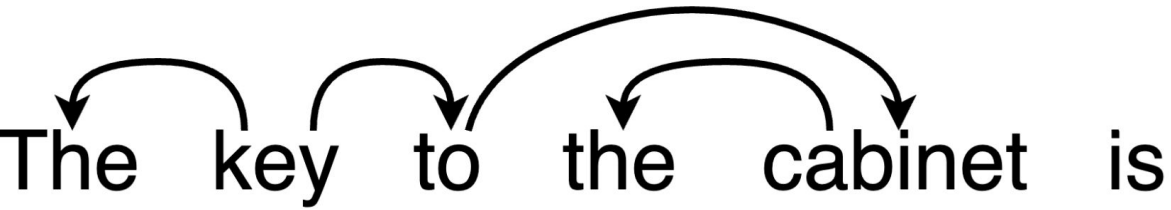

```
The
key
LA(key,the)
to
RA(key,to)
the
cabinet
LA(cabinet,the)
```

Transition-based dependency parsing (e.g. Nivre 2006)



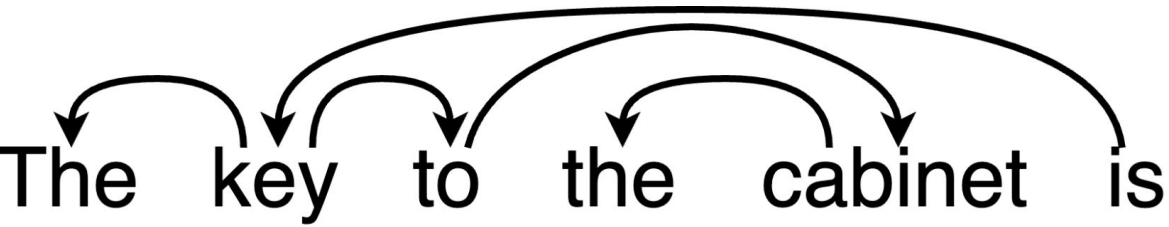

```
The
key
LA(key,the)
to
RA(key,to)
the
cabinet
LA(cabinet,the)
RA(to,cabinet)
```

Transition-based dependency parsing (e.g. Nivre 2006)



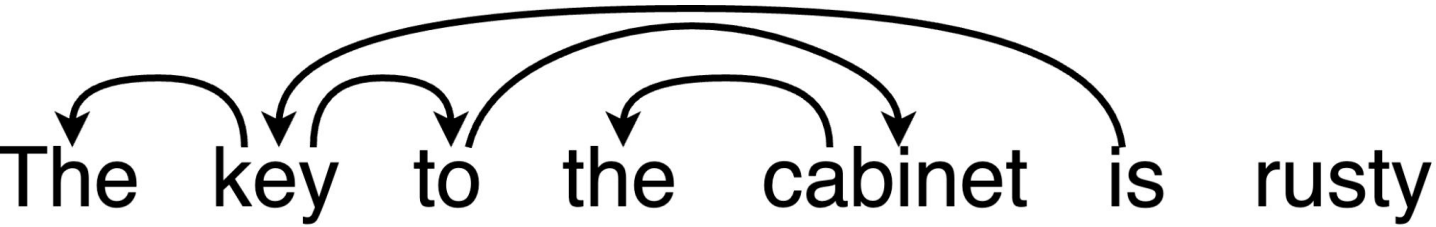

```
The
key
LA(key,the)
to
RA(key,to)
the
cabinet
LA(cabinet,the)
RA(to,cabinet)
is
```

Transition-based dependency parsing (e.g. Nivre 2006)



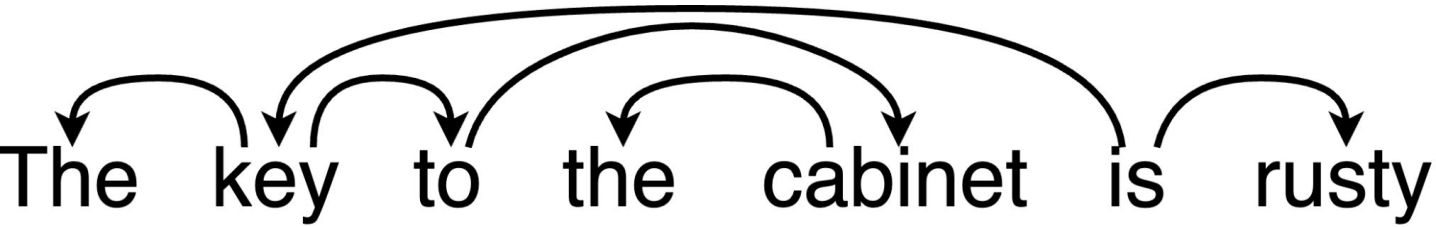

```
The
key
LA(key,the)
to
RA(key,to)
the
cabinet
LA(cabinet,the)
RA(to,cabinet)
is
LA(is,key)
```

Transition-based dependency parsing (e.g. Nivre 2006)



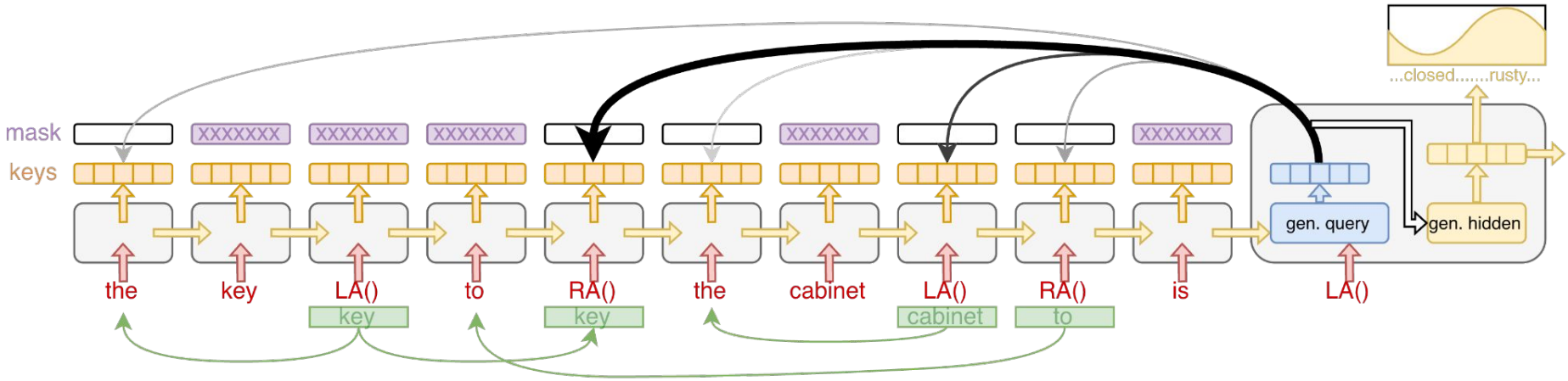

```
The
key
LA(key,the)
to
RA(key,to)
the
cabinet
LA(cabinet,the)
RA(to,cabinet)
is
LA(is,key)
rusty
```

Transition-based dependency parsing (e.g. Nivre 2006)

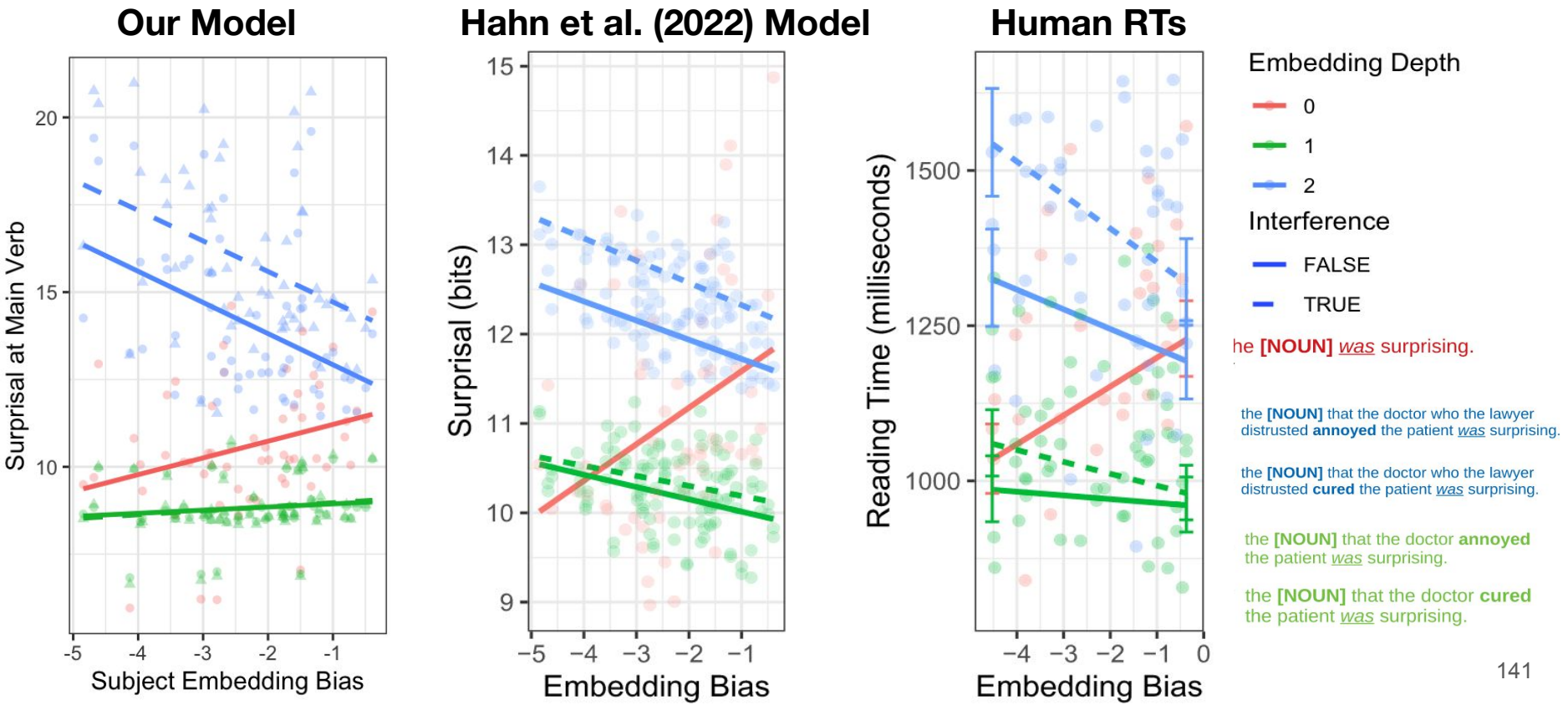



```
The
key
LA(key,the)
to
RA(key,to)
the
cabinet
LA(cabinet,the)
RA(to,cabinet)
is
LA(is,key)
rusty
RA(is,rusty)
```

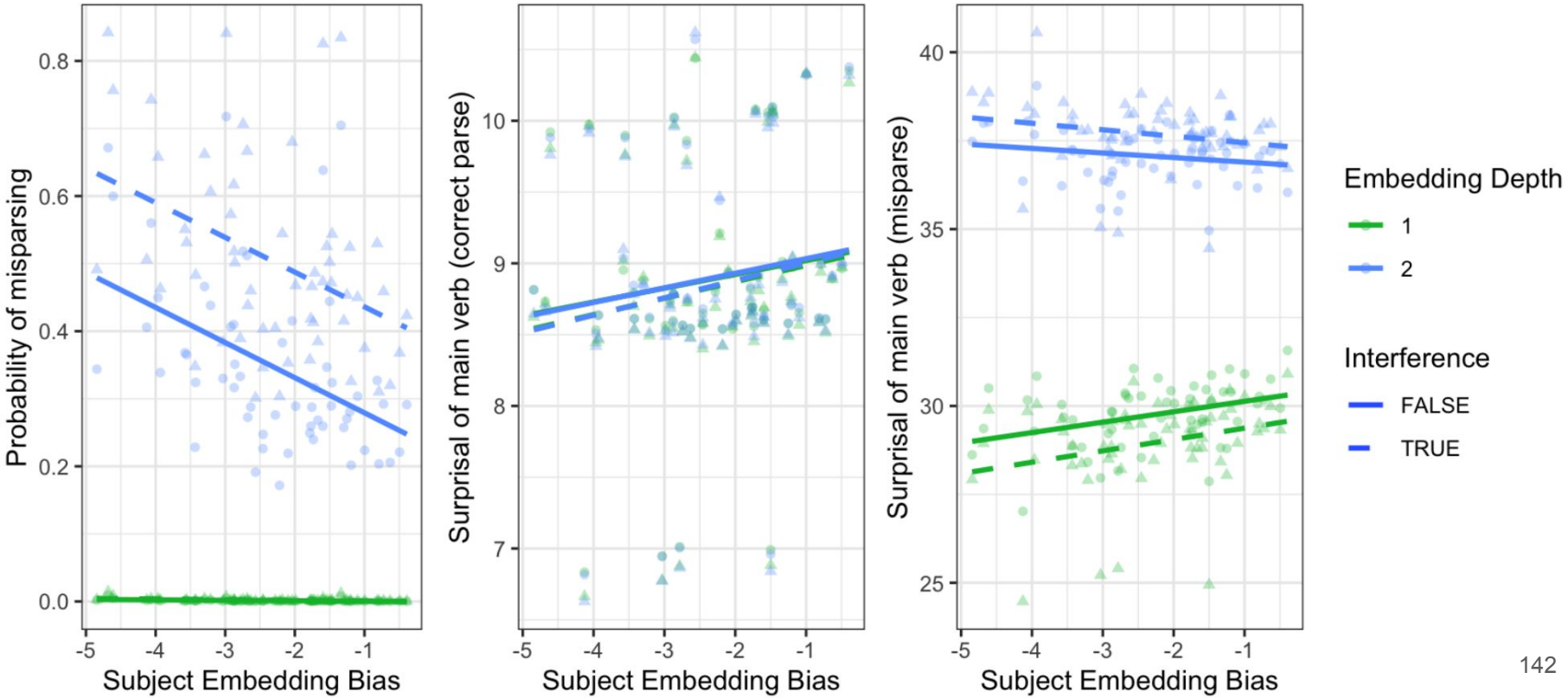
Model architecture



Detailed center embedding results



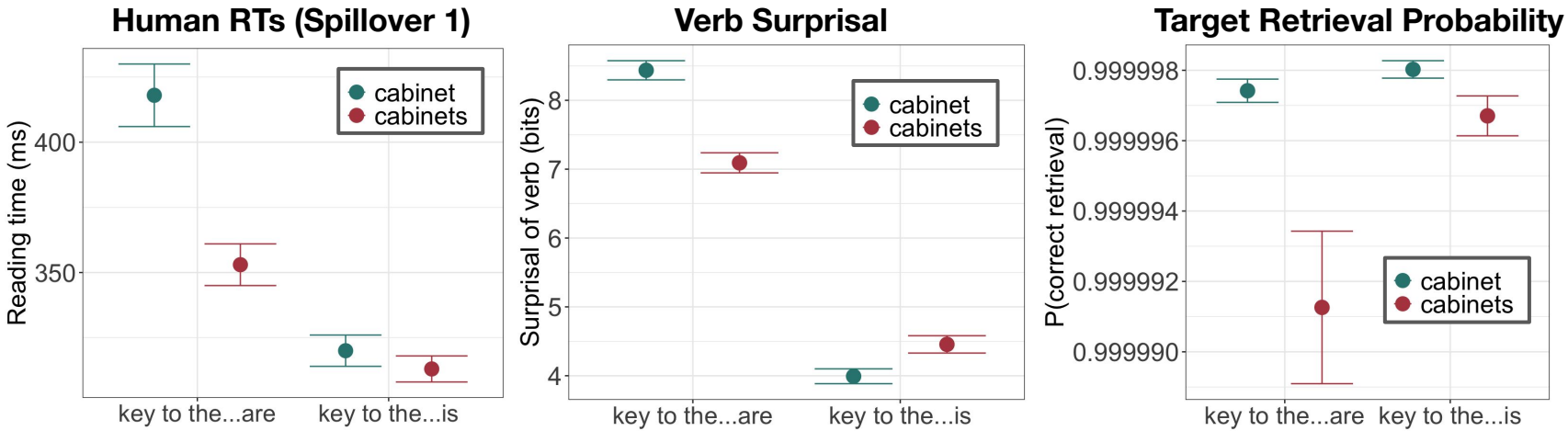
Detailed center embedding results



Full Wagers et al. (2009) Results

Wagers et al. (2009) Exp. 5

a) The key to the cabinet really are (ungrammatical, match)
b) The key to the cabinets really are (ungrammatical, mismatch)
c) The key to the cabinet really is (grammatical, match)
d) The key to the cabinets really is (grammatical, mismatch)



Direct manipulation of **[±has_head]**

a)	The *fact*	that the doctor who the nurse distrusted **annoyed**	the patient was…	(-resolved subject)
b)	The *fact* was	that the doctor who the nurse distrusted **annoyed**	the patient, and it was…	(+resolved subject)
c)	The *fact*	that the doctor who the nurse distrusted **cured**	the patient was…	(-resolved subject)
d)	The *fact* was	that the doctor who the nurse distrusted **cured**	the patient, and it was…	(+resolved subject)